

Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures

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Abstract

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1. Introduction

Directional representations arise whenever only the orientation of an observation is meaningful. Common examples include normalized document vectors, compositional profiles after normalization, and embedding vectors on the unit hypersphere. The von Mises–Fisher (vMF) distribution is a standard model for such data, and finite vMF mixtures provide a model-based alternative to spherical k-means (Banerjee et al., 2005). In high dimensions, however, the fitted component structure can involve many coordinates, making both estimation and interpretation difficult.

Sparse vMF mixtures have prominently been formulated through sparsity of the component direction vectors. Rossi and Barbaro (2022), for example, penalize their entries to estimate sparse directional prototypes. This target is appropriate when the scientific question concerns which coordinates describe each component prototype. It is not identical to asking which coordinates distinguish components in posterior classification. A coordinate can be active in every prototype and still contribute almost equally to all component scores. Conversely, a coordinate with a modest directional loading can be

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influential when component concentrations differ. The distinction is therefore between prototype support and the feature-dependent linear support of posterior component comparisons.

The construction draws on contrast penalties and grouped variable selection, which address related but distinct estimands. Penalized Gaussian mixtures have used entrywise mean shrinkage, grouped component means, and pairwise mean fusion (Pan and Shen, 2007; Xie et al., 2008; Guo et al., 2010). Sum-to-zero factor contrasts have also been used to separate common levels from heterogeneity (Bondell and Reich, 2009), while other directional-mixture methods target feature saliency, block structure, or hierarchical parameter regularization (Amayri and Bouguila, 2013; Salah et al., 2016; Gopal and Yang, 2014; Taghia et al., 2014; Salah and Nadif, 2019). Discriminative subspace approaches provide another route to variable relevance in model-based clustering (Bouveyron and Brunet, 2012; Bouveyron and Brunet-Saumard, 2014). These distinctions motivate connecting equality-based coordinate selection to posterior component comparison in a vMF mixture, where component-specific concentration prevents directional contrast alone from characterizing the relevant score contrast.

This paper targets posterior-score contrast support through natural parameters formed by multiplying each component direction by its concentration. For each observed coordinate, the component-specific natural parameters are decomposed into a common baseline and a centered contrast. The proposed centered natural-parameter group lasso (E-CGL) penalty groups the centered contrast across components, so its unit of regularization is the same coordinate-level unit used to define the target support. The common baseline remains unpenalized. An adaptive weighted extension and its path diagnostics are also considered. This adaptive centered natural-parameter group lasso (E-ACGL) is retained as a secondary model; E-CGL remains the primary estimator. Complete E-ACGL results and path-sensitivity analyses are provided in the Supplementary Material.

Candidate supports are generated by a guarded path with proximal-gradient updates inside a generalized expectation-maximization (GEM) algorithm. They are compared after support-constrained refitting that preserves the target. The primary selector is a practical Bayesian information criterion (BIC) computed from the refitted likelihood; alternative degrees of freedom and extended BIC (EBIC) rules are treated as sensitivity analyses. Support-recovery claims condition on a fixed number of components. When that number is unknown, component-number selection and support selection

are examined in separate stages rather than through a single joint criterion.

The empirical study has three parts. Oracle-error calibrated simulations compare posterior-score contrast support recovery across sample size, ambient dimension, and concentration heterogeneity. Shared-natural-parameter and dense-support designs separate the proposed estimand from prototype sparsity and record settings in which sparse posterior-score contrast support is inappropriate. A full-data Classic3 analysis examines clustering, coordinate reduction, and token-level contrasts on a vocabulary-aligned directional representation. Prespecified Classic3 splits assess support stability. Additional external-scope analyses are reported in the Supplementary Material.

The contributions are therefore: (i) a posterior-score contrast support estimand for vMF mixtures, its centered representation, and its decomposition into directional and concentration effects; (ii) E-CGL, together with guarded estimation and target-preserving refitting, with E-ACGL as a secondary fixed-pilot adaptive extension; (iii) existence on the bounded computational parameter space and conditional convergence results for the fixed-responsibility proximal problem and stationary accumulation points under stated regularity conditions; and (iv) an empirical evaluation that documents sparse, prototype, and dense-support regimes. E-CGL is not intended to replace sparse-prototype methods when prototype sparsity is the scientific target.

The remainder of the paper is organized as follows. Section 2 presents the vMF model, posterior-score contrast support, and centered group regularization. Section 3 develops the guarded proximal GEM estimator, pathwise support selection and refitting, and the structural and algorithmic properties used in the analysis. Sections 4 and 5 report the numerical studies and real-data analyses, respectively; the latter use prespecified preprocessing rules and split indices. Section 6 concludes with the empirical scope and limitations of the method.

2. Model and centered regularization

2.1. vMF mixture in natural parameters

Let $x_i \in \mathbb{S}^{d-1}$, $i = 1, \dots, n$, denote unit-normalized directional observations. A K -component vMF mixture has density

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\kappa_k) \exp\{\kappa_k \mu_k^\top x_i\},$$

where $\pi_k > 0$, $\sum_k \pi_k = 1$, $\|\mu_k\|_2 = 1$, and $\kappa_k \geq 0$. Its normalizing constant is

$$C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)}.$$

Here I_ν is the modified Bessel function of the first kind of order ν .

At $\kappa = 0$, this expression is interpreted by continuous extension as $C_d(0) = \Gamma(d/2)/(2\pi^{d/2})$.

Write

$$\begin{aligned} \Theta &= (\pi_1, \dots, \pi_K, \eta_1, \dots, \eta_K), \\ E &= (\eta_1, \dots, \eta_K)^\top, \quad M = (\mu_1, \dots, \mu_K)^\top, \end{aligned}$$

where $E, M \in \mathbb{R}^{K \times d}$ and $\eta_k = \kappa_k \mu_k \in \mathbb{R}^d$. Then

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i).$$

The component-level mapping back to (μ_k, κ_k) and its zero-concentration exception are recorded in Section 3.4. The observed log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^n \log \left\{ \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i) \right\},$$

where the component densities are defined with respect to surface-area measure on $\mathbb{S}^{d-1}$.

2.2. Posterior-score contrast support

Let $Z \in \{1, \dots, K\}$ denote the latent component label for a generic observation x . The component score

$$s_k(x) = \log \pi_k + \log C_d(\|\eta_k\|_2) + \eta_k^\top x$$

implies the pairwise posterior log-odds representation

$$\begin{aligned} \log \frac{\Pr(Z = k \mid x)}{\Pr(Z = \ell \mid x)} &= a_{k\ell} + (\eta_k - \eta_\ell)^\top x, \\ a_{k\ell} &= \log \frac{\pi_k}{\pi_\ell} + \log \frac{C_d(\|\eta_k\|_2)}{C_d(\|\eta_\ell\|_2)}. \end{aligned}$$

Thus $\eta_k - \eta_\ell$, rather than $\mu_k - \mu_\ell$, is the coefficient of the feature-dependent linear term. For coordinate j , write $E_{\cdot j} = \eta_{\cdot j}$ and let

$$H_K = I_K - K^{-1} \mathbf{1}_K \mathbf{1}_K^\top, \quad c_{\cdot j} = H_K \eta_{\cdot j} = \eta_{\cdot j} - \bar{\eta}_j \mathbf{1}_K,$$

where $\bar{\eta}_j = K^{-1} \sum_k \eta_{kj}$. We define

$$S_\eta = \{j : \|c_{\cdot j}\|_2 > 0\} = \bigcup_{k < \ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\}.$$

For population natural parameters $E^\star$, write $S_\eta^\star = \{j : \|H_K E_{\cdot j}^\star\|_2 > 0\}$.

We use *posterior-score contrast support* for this feature-dependent linear support. It does not include the intercepts $a_{k\ell}$. A coordinate with $c_{\cdot j} = 0$ cancels from every pairwise linear term, although its common baseline $\bar{\eta}_j$ remains in the fitted component densities and may affect $a_{k\ell}$ through the concentration norms. Proposition 1 gives the unique centered decomposition and its pairwise-dispersion identity.

For comparison, define

$$S_P = \{j : \|M_{\cdot j}\|_2 > 0\}, \quad S_\mu = \{j : \|H_K M_{\cdot j}\|_2 > 0\}.$$

The set S_P records coordinates present in at least one directional prototype, whereas S_μ records between-component variation in mean directions. The proposed S_η records coordinates entering posterior-score contrasts. The three targets can differ when component concentrations are heterogeneous.

Accordingly, $c_{\cdot j} \neq 0$ defines a posterior-score contrast coordinate; $c_{\cdot j} = 0$ with $\bar{\eta}_j \neq 0$ defines a common coordinate; and $c_{\cdot j} = 0$ with $\bar{\eta}_j = 0$ defines a globally null coordinate. These categories distinguish the proposed target from prototype support, which records nonzero component coordinates without requiring a difference between components.

The support is invariant to permutations of component labels because such a permutation only reorders the entries of $c_{\cdot j}$. It is not invariant to an arbitrary orthogonal rotation of the feature axes. As with other coordinate-selection procedures, its interpretation is therefore relative to the chosen coordinate system.

2.3. Centered natural-parameter group lasso (E-CGL)

The proposed E-CGL method regularizes the vMF natural parameters $\eta_k = \kappa_k \mu_k$, collected row-wise in $E \in \mathbb{R}^{K \times d}$. It penalizes the coordinate-wise contrasts $H_K E_{\cdot j}$ while leaving the common natural-parameter baseline unpenalized. For fixed K , the primary penalty functional is

$$\mathcal{P}_{\text{CGL}}(E) = \sum_{j=1}^d \|H_K E_{\cdot j}\|_2 = \sum_{j=1}^d \|c_{\cdot j}\|_2.$$

Each group is one observed coordinate across the K component contrasts, so the unit of regularization matches the definition of S_η . This coordinate-group construction follows the grouped-variable regularization principle of Yuan and Lin (2006). By contrast,

$$\|\eta_{\cdot j}\|_2^2 = K\bar{\eta}_j^2 + \|c_{\cdot j}\|_2^2,$$

so a raw- η group penalty combines common and contrast effects. A centered direction-space penalty targets S_μ , whereas the Rossi-style entrywise penalty targets S_P . When the κ_k differ, neither target is generally equal to the coefficient support of $\eta_k - \eta_\ell$ in posterior log-odds. These are distinct estimands, not competing parameterizations of the same support.

An entrywise centered penalty selects component-coordinate entries separately: $\lambda_\eta \sum_{k,j} |c_{kj}|$. E-CGL instead selects the complete coordinate contrast as one group. The resulting proximal map and label invariance are stated in Propositions 3 and 4.

2.4. Adaptive weighted extension (E-ACGL)

As a secondary extension, E-ACGL assigns a fixed pilot weight to each centered coordinate group. Let $a \in \mathcal{A} = \{\text{CGL}, \text{ACGL}\}$ denote the method and let E^{D} be the retained dense pilot. Define

$$w_j^{(a)} = \begin{cases} 1, & a = \text{CGL}, \\ \frac{\left(\|H_K E_{\cdot j}^{\text{D}}\|_2 + \epsilon\right)^{-\gamma}}{\text{median}_{1 \leq h \leq d} \left(\|H_K E_{\cdot h}^{\text{D}}\|_2 + \epsilon\right)^{-\gamma}}, & a = \text{ACGL}, \end{cases} \quad (1)$$

$$\mathcal{P}_a(E) = \sum_{j=1}^d w_j^{(a)} \|H_K E_{\cdot j}\|_2.$$

The dense pilot is shared by the two E-series methods, E-CGL and E-ACGL. The reported E-ACGL analyses use $\gamma = 1$ and $\epsilon = 10^{-6}$, and the weights are held fixed throughout optimization. Thus E-CGL is the special case $w_j^{(a)} = 1$ for every j , while both procedures target the same support S_η . Conditional on finite positive fixed weights, the centered proximal map and the fixed-responsibility convexity results below extend by replacing λ_η

with $\lambda_\eta w_j$ in group j . No adaptive oracle property or uniform performance advantage is claimed.

3. Estimation by guarded proximal generalized EM

3.1. Penalized objective

The finite vMF-mixture likelihood can diverge under concentration collapse (Ng, 2023). We therefore define the fitted model on

$$\mathcal{T}_{\kappa_{\max}} = \left\{ (\pi, E) : \pi \in [0, 1]^K, \mathbf{1}_K^\top \pi = 1, \|\eta_k\|_2 \leq \kappa_{\max} \text{ for every } k \right\},$$

with $\kappa_{\max} = 10^6$ in all reported analyses. The numerical mixing-weight floor is an implementation safeguard; the existence argument below uses the closed simplex, and the local convergence statement assumes the floor and the concentration bound are inactive.

For fixed K , λ_η , and $a \in \mathcal{A}$, the path-generating criterion is

$$\begin{aligned} \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta) &= \ell(\Theta) - \lambda_\eta \mathcal{P}_a(E) \\ &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^d w_j^{(a)} \|H_K \eta_j\|_2. \end{aligned} \tag{2}$$

The penalty and the vMF normalizing constant are coupled through η_k , so the natural-parameter update is computed by a guarded proximal-gradient generalized M-step.

Lemma 1 (Existence on the finite parameter space). *For fixed K, d, n , finite $\kappa_{\max}$, nonnegative λ_η , and finite positive fixed weights, the criterion in (2) attains a maximum on $\mathcal{T}_{\kappa_{\max}}$.*

The lemma establishes existence for the bounded computational problem. It does not imply uniqueness, mixture identifiability, or attainment of a global maximum by the numerical algorithm.

3.2. Fixed-penalty guarded proximal GEM

We use a generalized expectation-maximization (GEM) procedure at each fixed value of λ_η . The E-step computes the posterior responsibilities exactly. In the generalized M-step, the mixing proportions are updated by (8), while the natural-parameter block receives finitely many safeguarded proximal-gradient updates. This block is required to increase the penalized auxiliary

criterion within numerical tolerance rather than to attain its exact conditional maximum. The procedure is therefore a GEM algorithm rather than an exact EM algorithm.

3.2.1. *E-step and fixed-responsibility subproblem*

At outer iteration t , compute $\tau_{ik}^{(t)}$ and

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} C_d(\|\eta_k^{(t)}\|_2) \exp\{(\eta_k^{(t)})^\top x_i\}}{\sum_{h=1}^K \pi_h^{(t)} C_d(\|\eta_h^{(t)}\|_2) \exp\{(\eta_h^{(t)})^\top x_i\}}, \quad (3)$$

and define

$$N_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)}, \quad r_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)} x_i.$$

Conditional on these responsibilities and the fixed weights for method a , the cap-inactive natural-parameter block is represented by

$$\begin{aligned} \min_E \quad & F_t^{(a)}(E) = f_t(E) + g_{\lambda_\eta}^{(a)}(E), \\ & f_t(E) = - \sum_{k=1}^K \left\{ (r_k^{(t)})^\top \eta_k + N_k^{(t)} \log C_d(\|\eta_k\|_2) \right\}, \\ & g_{\lambda_\eta}^{(a)}(E) = \lambda_\eta \sum_{j=1}^d w_j^{(a)} \|H_K \eta_j\|_2. \end{aligned} \quad (4)$$

The numerical algorithm evaluates this objective through cap-feasible trial iterates. It rejects rather than projects a proposal that violates the concentration cap; the cap-active constrained subproblem is not claimed to be solved by the closed-form proximal update.

The smooth term is convex because $-\log C_d(\|\eta_k\|_2)$ is the vMF log-partition function. Its gradient is

$$\nabla_{\eta_k} f_t(E) = N_k^{(t)} A_d(\|\eta_k\|_2) \frac{\eta_k}{\|\eta_k\|_2} - r_k^{(t)}, \quad (5)$$

with the continuous value $-r_k^{(t)}$ at $\eta_k = 0$, where

$$A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}$$

is the vMF mean-resultant-length function, with $A_d(0) = 0$ by continuous extension.

3.2.2. Generalized M-step and centered proximal update

Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For a step size $s > 0$, set

$$V = E^{(u)} - s\nabla f_t(E^{(u)}). \quad (6)$$

The update separates by observed coordinate and has the closed form

$$E_{\cdot j}^{(u+1)} = J_K V_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j^{(a)}}{\|H_K V_{\cdot j}\|_2}\right)_+ H_K V_{\cdot j}. \quad (7)$$

Here $(z)_+ = \max(z, 0)$, and the second term is defined as zero when $H_K V_{\cdot j} = 0$.

The common component is therefore retained, while the centered contrast is group-thresholded; this is a proximal group soft-thresholding operation (Parikh and Boyd, 2014). Mixing proportions are updated by

$$\pi_k^{\text{cand}} = \frac{\max(N_k^{(t)}/n, 10^{-12})}{\sum_h \max(N_h^{(t)}/n, 10^{-12})}. \quad (8)$$

The initial step size is $s = 1/\max_k N_k^{(t)}$. If the smooth majorization condition

$$f_t(E^{(u+1)}) \leq f_t(E^{(u)}) + \langle \nabla f_t(E^{(u)}), E^{(u+1)} - E^{(u)} \rangle + \frac{\|E^{(u+1)} - E^{(u)}\|_F^2}{2s} \quad (9)$$

is not satisfied within numerical tolerance, s is halved and the proximal proposal is recomputed. This backtracking is repeated for at most 40 trials.

Equivalently, define the majorization residual

$$R_{\text{maj}}(E^+, E; s) = f_t(E^+) - f_t(E) - \langle \nabla f_t(E), E^+ - E \rangle - \frac{\|E^+ - E\|_F^2}{2s}. \quad (10)$$

The numerical update is accepted only when $R_{\text{maj}}(E^+, E; s) \leq \varepsilon_{\text{maj}}$. A proposal with $\max_k \|\eta_k^+\|_2 > \kappa_{\text{max}}$ is not projected in this proximal loop; instead, the step is halved and the proposal is recomputed. If no acceptable proposal is found in 40 trials, the current path point is marked invalid and the remaining single path is not evaluated.

3.2.3. Safeguards and stopping rules

Let $Q(\Theta \mid \tau)$ be the expected complete-data log-likelihood. For an old iterate Θ and candidate Θ^+ , define

$$\begin{aligned} Q_{\lambda_\eta}^{(a)}(\Theta \mid \tau) &= Q(\Theta \mid \tau) - \lambda_\eta \mathcal{P}_a(E), \\ \Delta Q_{\lambda_\eta}^{(a)} &= Q_{\lambda_\eta}^{(a)}(\Theta^+ \mid \tau) - Q_{\lambda_\eta}^{(a)}(\Theta \mid \tau), \\ \Delta \mathcal{L}_{\lambda_\eta}^{(a)} &= \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^+) - \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta). \end{aligned} \tag{11}$$

The guarded generalized M-step accepts the candidate only if

$$\Delta Q_{\lambda_\eta}^{(a)} \geq -\varepsilon_{\text{acc}}, \quad \Delta \mathcal{L}_{\lambda_\eta}^{(a)} \geq -\varepsilon_{\text{acc}}. \tag{12}$$

Otherwise, the path fit returns the previous accepted estimate only with a failure diagnostic; its support is not recorded and the remaining path is not evaluated.

Components with effective count below 10^{-8} are retained to preserve the fixed value of K and the event is recorded. They are not pruned or reinitialized. These safeguards support statements about accepted numerical updates; they do not establish convergence of the complete parameter sequence, stationarity of every fitted path point, or global optimality.

The recorded analyses use the absolute support threshold $\varepsilon_{\text{supp}} = 10^{-8}$. This threshold is a numerical convention for reading a fitted path support, not an additional penalty parameter.

The recorded simulation and real-data controls use at most 200 and 160 proximal-gradient iterations per generalized M-step, respectively. The inner and outer stopping rules are

$$\frac{\|E^{(u+1)} - E^{(u)}\|_F}{\max\{1, \|E^{(u)}\|_F\}} < \varepsilon_{\text{PG}}, \tag{13}$$

$$\frac{|\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t+1)}) - \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)})|}{\max\{1, |\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)})|\}} < \varepsilon_{\text{conv}}. \tag{14}$$

The numerical tolerances are

$$(\varepsilon_{\text{PG}}, \varepsilon_{\text{maj}}, \varepsilon_{\text{acc}}, \varepsilon_{\text{conv}}, \varepsilon_{\text{supp}}) = (10^{-8}, 10^{-10}, 10^{-8}, 10^{-7}, 10^{-8}).$$

The implementation permits at most 40 backtracking attempts and 100 outer GEM iterations.

3.3. Pathwise selection and support-constrained refit

3.3.1. Method-specific zero-augmented Karush–Kuhn–Tucker-scaled path

Let $\hat{\Theta}^D$ be the numerically admissible dense fit with the largest finite observed log-likelihood among the nonfailed multistart fits. Convergence status is retained as a diagnostic but is not required for dense-start admissibility. After the weights in (1) are computed and held fixed, the implementation first obtains the guarded zero-penalty fit

$$\hat{\Theta}_0^{(a)} = \text{Fit}_{\lambda_\eta=0}^{(a)} \left(\hat{\Theta}^D; w^{(a)} \right).$$

Responsibilities τ^0 under $\hat{\Theta}_0^{(a)}$ define (N_k^0, r_k^0) for path construction. Let $r_\bullet = \sum_k r_k^0$ and $N_\bullet = \sum_k N_k^0 = n$, and let η^{common} be the pooled finite-cap vMF estimate formed from $(N_\bullet, r_\bullet)$. Define $E_0^{\text{com}} = \mathbf{1}_K(\eta^{\text{common}})^\top$ and $G_0 = \nabla f_{\tau^0}(E_0^{\text{com}})$. A Karush–Kuhn–Tucker (KKT)-scale upper endpoint for this initial fixed-responsibility subproblem is

$$\tilde{\lambda}_{\max, \eta}^{(a)} = 1.05 \max_{1 \leq j \leq d} \frac{\|(H_K G_0)_{\cdot j}\|_2}{w_j^{(a)}}. \quad (15)$$

This endpoint is a scale for the zero-penalty-responsibility convex subproblem, not a global threshold for the observed mixture objective. The single path for method a is

$$\Lambda^{(a)} = \{0\} \cup \left\{ \tilde{\lambda}_{\max, \eta}^{(a)} 10^{-4+4q/(L-2)} : q = 0, \dots, L-2 \right\}, \quad (16)$$

with $L = 240$. Thus the 239 positive values range from $10^{-4}\tilde{\lambda}_{\max, \eta}^{(a)}$ to $\tilde{\lambda}_{\max, \eta}^{(a)}$ on a logarithmic grid. E-CGL and E-ACGL use the same relative range and grid size; their absolute penalty values differ because the KKT endpoint contains the fixed method-specific weights. Warm starts are used in increasing penalty order.

Let

$$\mathcal{I}_{\text{record}}^{(a)} = \left\{ \ell : \hat{\Theta}_\ell^{(a)} \text{ is finite, nonfailed, and cap-feasible} \right\}.$$

A finite path fit that reaches an iteration limit is stored with **converged** = **FALSE**; a numerical failure is not stored and terminates the remaining path. This distinction prevents a failed update from being reported as convergence. The implementation stops early after storing an active support with at most one coordinate. Independently of this stopping rule, the all-common model

is added explicitly as the empty support. The tuning values index candidate supports; the final inferential output is the selected support and its penalty-free constrained refit. The early stop is a computational heuristic and does not assert nested supports along the nonconvex mixture path. Consequently, every nonempty selection is conditional on the finite, recorded, and possibly truncated path that was actually evaluated.

3.3.2. Target-preserving support-constrained refit

The penalized path identifies candidate supports, but the final estimates are obtained by a sparsity-penalty-free, finite-cap constrained refit. For a candidate support S , use the parameterization

$$\eta_{kj} = b_j + c_{kj}, \quad \sum_{k=1}^K c_{kj} = 0, \quad c_{kj} = 0 \quad (j \notin S). \quad (17)$$

Hence $j \notin S$ implies $\eta_{1j} = \dots = \eta_{Kj} = b_j$: an inactive posterior-score contrast coordinate retains its estimated common baseline in the natural-parameter space.

For each active coordinate, the first $K - 1$ contrasts are free and the last is set to their negative sum. Conditional on the responsibilities, the natural-parameter update targets maximization, over $b \in \mathbb{R}^d$ and the free active contrasts, of

$$Q_S(\eta) = \sum_{k=1}^K \left[\eta_k^\top r_k + N_k \log C_d(\|\eta_k\|_2) \right] \quad (18)$$

The implementation projects the recorded path fit that generated the candidate support onto the support constraint and, when needed, scales the complete natural-parameter matrix to the concentration cap before optimization. The smooth conditional M-step in this refit is solved using the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm with bounds (L-BFGS-B) (Byrd et al., 1995), with analytic gradients consistent with the numerically evaluated vMF normalizer. This quasi-Newton routine is used only in the penalty-free refit; the penalized E-series update remains the centered proximal-gradient generalized M-step. Although L-BFGS-B supports coordinatewise bounds, the row-norm restriction $\|\eta_k\|_2 \leq \kappa_{\max}$ is not a coordinatewise box constraint. The update from the old iterate toward the L-BFGS-B proposal is therefore step-halved until both the cap and the observed log-likelihood guard hold. Responsibilities, mixing proportions, and

the constrained natural parameters are iterated until the numerical stopping criterion is met.

The recorded refit uses at most 160 outer iterations followed, when the first fit is finite and nonfailed but unconverged, by at most 440 additional iterations. Each L-BFGS-B M-step uses at most 80 iterations and projected-gradient tolerance 10^{-6} . The outer refit stops when the absolute log-likelihood increment is no larger than $10^{-7} + 10^{-8}|\ell_{\text{old}}|$.

A refit is eligible for BIC comparison only when it is finite, converged, nonfailed, has a successful inner optimizer, satisfies the centered-support constraint to numerical tolerance, and has $\max_k \hat{\kappa}_k < 0.99\kappa_{\text{max}}$. The explicit empty-support refit is initialized from the recorded fit with the smallest support; ties use the largest observed log-likelihood and then the fixed path order.

The target-preserving support-constrained refit enforces the centered support constraint directly while retaining the common baseline. The constrained likelihood is optimized numerically and is not claimed to attain the global maximum of the nonconvex mixture likelihood.

The earlier active-only refit imposed $\eta_j = 0$ outside S and therefore removed both the centered contrast and the common baseline. It is retained only as an ablation diagnostic; the target-preserving support-constrained refit is used in the primary analyses.

3.3.3. Support selection by BIC after refit

Let $m_S = |S|$. The practical degrees-of-freedom approximation is

$$\text{df}(S) = d + (K - 1)m_S + (K - 1)\mathbf{1}(m_S > 0). \quad (19)$$

The three terms count the coordinatewise common baselines, the active centered contrasts, and the mixing proportions when nonzero component contrasts are present. This generic nominal dimension is a practical model-selection approximation, not an exact effective degrees of freedom for a penalized nonregular mixture.

For each recorded path point, define

$$\begin{aligned}
S_\ell^{(a)} &= \left\{ j : \|H_K \hat{E}_{\ell,j}^{(a)}\|_2 > \varepsilon_{\text{supp}} \right\}, \\
\mathcal{S}_{\text{path}}^{(a)} &= \text{ordered unique } \left\{ S_\ell^{(a)} : \ell \in \mathcal{I}_{\text{record}}^{(a)} \right\}, \\
\mathcal{S}_{\text{cand}}^{(a)} &= \mathcal{S}_{\text{path}}^{(a)} \cup \{\emptyset\}, \\
\mathcal{S}_{\text{elig}}^{(a)} &= \left\{ S \in \mathcal{S}_{\text{cand}}^{(a)} : \hat{\Theta}_S^{\text{refit}} \text{ passes all numerical checks} \right\}.
\end{aligned} \tag{20}$$

Candidate construction, refit scoring, and final selection are then

$$\begin{aligned}
\text{BIC}^{\text{refit}}(S) &= -2\ell(\hat{\Theta}_S^{\text{refit}}) + \log(n) \text{df}(S), \\
\hat{S}^{(a)} &= \arg \min_{S \in \mathcal{S}_{\text{elig}}^{(a)}} \text{BIC}^{\text{refit}}(S).
\end{aligned} \tag{21}$$

The all-common candidate is always refitted and enters the BIC comparison if eligible. Nonempty support candidates remain conditional on the finite path. For a nonempty population support, the event

$$S_\eta^* \in \mathcal{S}_{\text{path}}^{(a)}$$

defines path coverage. In simulations we separately report path coverage, overall exact recovery, and

$$\Pr \left(\hat{S}^{(a)} = S_\eta^* \mid S_\eta^* \in \mathcal{S}_{\text{path}}^{(a)} \right),$$

the conditional BIC success rate. When $S_\eta^* = \emptyset$, candidate coverage equals one by construction, and exact recovery is the event $\hat{S}^{(a)} = \emptyset$. Adding the empty support independently of the recorded path prevents the stopping rule from omitting the all-common model. An audit of the previously recorded paths showed that this addition left all primary and estimand-separation conclusions unchanged, but selected the empty support in a subset of stress-condition repetitions; those stress summaries were recomputed under the corrected candidate family.

When several recorded path fits generate the same support, the fit with the largest observed log-likelihood initializes the single constrained refit. This rule is applied before support-level BIC comparison and does not use the external class labels. The explicitly appended empty support uses the smallest-support, largest-log-likelihood, fixed-order initialization rule stated above. Exact BIC ties are resolved by the first eligible candidate in the fixed candidate order; this deterministic convention has no inferential meaning.

The tuning parameter is therefore treated only as a candidate-generation index. BIC after refit selects a distinct support and does not define or require a uniquely interpreted estimate of λ_η .

All distinct nonempty supports encountered on the path and the explicitly added empty support are refitted. The information criterion follows the BIC principle of Schwarz (1978), using the generic nominal dimension above. It is a within-method support selector, not an exact marginal-likelihood approximation or a consistency result for a singular mixture family.

Each competing procedure has its own support target and nominal dimension. BIC therefore selects refitted candidates separately within each method. Values across methods are descriptive and do not provide a common-scale ranking of the support targets.

An EBIC sensitivity criterion is

$$\text{EBIC}_\gamma(S) = \text{BIC}^{\text{refit}}(S) + 2\gamma \log \binom{d}{m_S}, \quad \gamma \in \{0.25, 0.5, 1\}.$$

EBIC (Chen and Chen, 2008) and alternative degrees-of-freedom rules are treated as sensitivity analyses rather than changes to the primary selector.

Algorithm 1 writes out the E-step, the guarded proximal generalized M-step, path construction, and refit-based support selection. E-CGL is obtained with unit weights; E-ACGL uses the fixed dense-pilot weights in (1). All symbols and updates in the listing are defined in the preceding subsections. The fixed- λ subroutine returns **FAIL** only after a numerical guard or finite-value check fails. Reaching an iteration limit returns a nonfailed fit with its convergence flag set to false, so that failure and nonconvergence remain distinct. Lines 4–17 of the listing define the fixed- λ_η subroutine $\text{Fit}_{\lambda_\eta}^{(a)}$ used in the subsequent path steps.

Algorithm 1. Guarded proximal GEM and support selection

Require: row-normalized $X \in \mathbb{R}^{n \times d}$; K ; $a \in \mathcal{A}$; dense starts; $L = 240$; $\kappa_{\max}$; iteration limits $T_{\max}$ and $U_{\max}$; the stopping tolerances in Eqs. (13)–(14); ε_{maj} , ε_{acc} , $\varepsilon_{\text{supp}}$; and the backtracking limit.

Ensure: $\hat{S}^{(a)}$ and $\hat{\Theta}_{\hat{S}^{(a)}}^{\text{refit}}$.

- 1: **Step 1: Initialization.**
 - 2: **(a) Dense pilot:** retain the admissible dense multistart fit with the largest finite observed log-likelihood as $\hat{\Theta}^D$.
 - 3: **(b) Fixed weights:** compute and fix $w_j^{(a)}$, $j = 1, \dots, d$, according to Eq. (1).
 - 4: **Step 2: Fixed- λ_η proximal GEM.**
 - 5: Initialize $\Theta^{(0)}$ from the supplied warm start and set **converged** = FALSE.
 - 6: **for** $t = 0, \dots, T_{\max} - 1$ **do**
 - 7: **Step 2.1: E-step.** Compute $\tau_{ik}^{(t)}$, $N_k^{(t)}$, and $r_k^{(t)}$ using Eq. (3) and the definitions immediately below it.
 - 8: **Step 2.2: Update mixing proportions.** Compute π^{cand} by Eq. (8).
 - 9: **Step 2.3: Update natural parameters.** Initialize the inner iteration at the current natural-parameter matrix.
 - 10: **for** $u = 0, \dots, U_{\max} - 1$ **do**
 - 11: $s \leftarrow (\max_k N_k^{(t)})^{-1}$, $V \leftarrow E^{(u)} - s \nabla f_t(E^{(u)})$.
 - 12: $E_{\cdot j}^{(u+1)} \leftarrow J_K V_{\cdot j} + \left(1 - \frac{s \lambda_\eta w_j^{(a)}}{\|H_K V_{\cdot j}\|_2}\right) H_K V_{\cdot j}$, $j = 1, \dots, d$.
 - 13: Halve s and recompute V and $E^{(u+1)}$ until the majorization and concentration-cap conditions hold; if no admissible proposal is found, **return FAIL**.
 - 14: If Eq. (13) holds, **break**; **end for**; set $\Theta^+ \leftarrow (\pi^{\text{cand}}, E^{(u+1)})$.
 - 15: **Step 2.4: Acceptance and convergence.** Accept Θ^+ only if Eq. (12) holds; otherwise **return FAIL**.
 - 16: Set $\Theta^{(t+1)} \leftarrow \Theta^+$; if Eq. (14) holds, set **converged** = TRUE and **break**.
 - 17: **end for**; at the iteration limit, retain the current finite fit with **converged** = FALSE.
 - 18: **Step 3: Increasing-penalty path.**
 - 19: $\hat{\Theta}_0^{(a)} \leftarrow \text{Fit}_{\lambda_\eta=0}^{(a)}(\hat{\Theta}^D; w^{(a)})$; $\hat{\Theta}_0^{(a)} = \text{FAIL} \Rightarrow \text{return FAIL}$.
 - 20: Set $\mathcal{I}_{\text{record}}^{(a)} \leftarrow \{0\}$ and form $S_0^{(a)}$ from $\hat{\Theta}_0^{(a)}$ by Eq. (20).
 - 21: Construct $\hat{\lambda}_{\max, \eta}^{(a)}$ and the single path $\Lambda^{(a)}$ from Eqs. (15)–(16).
 - 22: **for** each positive $\lambda_{\eta, \ell} \in \Lambda^{(a)}$ in increasing order **do** initialize Step 2 from the preceding accepted fit.
 - 23: Repeat Step 2. If the fit is finite, nonfailed, and cap-feasible, include ℓ in $\mathcal{I}_{\text{record}}^{(a)}$ and form $S_\ell^{(a)}$ by Eq. (20).
 - 24: Numerical failure terminates the remaining path; after recording $|S_\ell^{(a)}| \leq 1$, stop the path. **end for**
 - 25: **Step 4: Target-preserving refit and support selection.**
 - 26: Construct $\mathcal{S}_{\text{cand}}^{(a)}$ by Eq. (20); for each candidate, compute $\hat{\Theta}_S^{\text{refit}}$ under Eq. (17), apply the stated numerical checks, and then form $\mathcal{S}_{\text{elig}}^{(a)}$.
 - 27: Select $\hat{S}^{(a)}$ by Eq. (21); if $\mathcal{S}_{\text{elig}}^{(a)} = \emptyset$, **return FAIL**.
 - 28: **return** $(\hat{S}^{(a)}, \hat{\Theta}_{\hat{S}^{(a)}}^{\text{refit}})$.
-

In the proximal update, the centered term is zero when $H_K V_{\cdot j} = 0$. The guard logic is common to both methods, whereas the penalized criteria evaluated by the guards are method-specific through $w_j^{(a)}$. The refit is a numerical constrained alternating procedure, not a claim of a global constrained maxi-

mum. Refit BIC compares eligible supports only within a method.

3.3.4. Selection of the number of components

Component-number selection is not part of the E-CGL support estimator. The number of components and the sparsity level are therefore selected in separate stages. In support-recovery simulations, K is fixed at the data-generating value so that support selection is not confounded with component-number misspecification.

For an unknown K , let $\mathcal{K}$ be a prespecified candidate set. The practical two-stage procedure is:

1. Fit an unpenalized dense vMF mixture for each $K \in \mathcal{K}$ using multiple starts.
2. Compare candidate K values using label-free density criteria, information criteria, and partition stability as appropriate to the application.
3. Fix the selected component resolution $\widehat{K}$.
4. Fit the E-CGL path at $\widehat{K}$ and choose a distinct support by BIC after the target-preserving support-constrained refit.

Symbolically,

$$\widehat{K} = \arg \min_{K \in \mathcal{K}} \mathcal{C}_{\text{dense}}(K), \quad \widehat{S}_{\widehat{K}} = \arg \min_{S \in \mathcal{S}_{\text{cand}}(\widehat{K})} \text{BIC}^{\text{refit}}(\widehat{K}, S).$$

Here $\mathcal{C}_{\text{dense}}$ may use independent or held-out negative log-likelihood, bootstrap out-of-bag negative log-likelihood, or an information criterion. No single information criterion is asserted to select the correct resolution in every high-dimensional mixture. When density fit and stability favor different K , both competing resolutions are reported.

3.3.5. Computational implementation

The implementation uses log-sum-exp stabilization in the E-step, warm starts along the penalty path, and explicit convergence and constraint checks in the centered-support refit. Optional compiled helpers implemented through the Rcpp interface (Eddelbuettel and François, 2011) replace low-level numerical operations without changing the statistical algorithm; R-only and Rcpp-helper outputs were checked to numerical tolerance before runtime comparisons.

The current computational claims are limited to empirical equality and runtime diagnostics. The method does not rely on an Rcpp implementation for its statistical definition.

The finite parameter space $\mathcal{T}_{\kappa_{\max}}$ is used in dense initialization, the penalized path, and support-constrained refitting. Backtracking rejects any proposed E-series iterate outside this space. A converged refit that reaches 99% of the bound is retained for diagnosis but is not eligible for information-criterion selection. This guard defines the finite computational problem; it is not an additional sparsity penalty. Target-preserving refitting refers to enforcement of the centered-support constraint, not to a global mixture optimum.

3.4. Structural properties

The following statements justify the centered support estimand for a fixed number of components. They do not imply uniqueness of a fitted mixture, which remains subject to finite-mixture identifiability and label permutation. Full proofs are given in Appendix Appendix A.

Proposition 1 (Centered decomposition and pairwise support). *For any $v \in \mathbb{R}^K$, there is a unique decomposition*

$$v = \bar{v}\mathbf{1}_K + c, \quad \bar{v} = K^{-1}\mathbf{1}_K^\top v, \quad \mathbf{1}_K^\top c = 0.$$

For $v = \eta_{\cdot j}$, the following statements are equivalent:

$$c = 0; \quad \eta_{1j} = \cdots = \eta_{Kj}; \quad \eta_{kj} - \eta_{\ell j} = 0 \quad \text{for all } k < \ell.$$

Moreover,

$$\|c\|_2^2 = \frac{1}{K} \sum_{k < \ell} (\eta_{kj} - \eta_{\ell j})^2.$$

Consequently,

$$S_\eta = \bigcup_{k < \ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\},$$

the union of the feature-dependent linear supports in all pairwise posterior log-odds.

Proposition 1 clarifies the scope of the estimand. If $c_{.j} = 0$, the coefficient of x_j cancels from every pairwise posterior log-odds. The common baseline is nevertheless retained in η_k and may affect the intercept through $\|\eta_k\|_2$; it is therefore not irrelevant to the component density.

Proposition 2 (Natural-parameter and directional decomposition). *Let $\kappa = (\kappa_1, \dots, \kappa_K)^\top$, $\bar{\kappa} = K^{-1}\mathbf{1}_K^\top \kappa$, and $h = \kappa - \bar{\kappa}\mathbf{1}_K$. Write*

$$\bar{\mu} = K^{-1}M^\top \mathbf{1}_K, \quad U = H_K M.$$

Then $M = \mathbf{1}_K \bar{\mu}^\top + U$ and

$$H_K E = \bar{\kappa} U + h \bar{\mu}^\top + H_K \text{diag}(h) U.$$

The three terms represent directional heterogeneity, concentration heterogeneity along the mean directional loading, and their interaction.

If $\kappa_1 = \dots = \kappa_K = \kappa > 0$, then $H_K E = \kappa H_K M$ and $S_\eta = S_\mu$. If $\mu_1 = \dots = \mu_K = \bar{\mu}$ and $h \neq 0$, then $S_\mu = \emptyset$ while $S_\eta = \text{supp}(\bar{\mu})$. Conversely, heterogeneous concentrations can make $H_K M_{.j} \neq 0$ even when $H_K E_{.j} = 0$. In the general decomposition the three terms may cancel, so S_η is not identified with the union of their separate supports.

Proposition 3 (Label equivariance and support invariance). *Let P be a $K \times K$ permutation matrix. Then*

$$H_K P = P H_K, \quad \|H_K P \eta_{.j}\|_2 = \|H_K \eta_{.j}\|_2.$$

The E-CGL penalty is invariant to component relabeling, its proximal map is equivariant, and S_η is unchanged.

For a nonzero natural parameter, the component representation is recovered as $\kappa_k = \|\eta_k\|_2$ and $\mu_k = \eta_k / \|\eta_k\|_2$. At $\eta_k = 0$, the vMF component is uniform and its direction is not identified. These algebraic statements do not establish finite-mixture identifiability or uniqueness of the fitted mixture. Component-specific interpretation assumes distinct component distributions and positive mixing weights and is made only up to label permutation. If two components have identical natural parameters, their individual mixing weights are not separately identified. Support-recovery statements are conditional on fixed K .

3.5. Algorithmic properties

Proposition 4 (Closed-form proximal update). *Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For $\xi \geq 0$ and $v \in \mathbb{R}^K$,*

$$\text{prox}_{\xi\|H_K\cdot\|_2}(v) = J_K v + \left(1 - \frac{\xi}{\|H_K v\|_2}\right)_+ H_K v,$$

where the second term is defined as zero when $H_K v = 0$.

The common baseline is unchanged. Group soft thresholding is applied only to the centered contrast, matching the coordinate-level definition of S_η .

Proposition 5 (Smooth majorization and proximal descent). *Fix responsibilities with $N_k > 0$. The function f_t in Section 3.2.1 is convex and has a Lipschitz gradient with constant $L \leq \max_k N_k$. Hence, for $s \leq 1/L$, the proximal-gradient update in Section 3.2.2 does not increase $F_t^{(a)}$ in exact arithmetic. More generally, an update accepted with majorization tolerance ε_{maj} satisfies*

$$F_t^{(a)}(E^{(u+1)}) \leq F_t^{(a)}(E^{(u)}) + \varepsilon_{\text{maj}}.$$

Proposition 5 motivates the backtracking rule. The implementation starts from $s = 1/\max_k N_k$ and halves the step until the majorization inequality is satisfied numerically. This check also protects against error from numerical evaluation of the vMF normalizing constant.

Theorem 1 (Fixed-responsibility proximal convergence). *Let $d \geq 2$. Fix responsibilities such that $N_k > 0$ and $\|r_k\|_2 < N_k$ for every k , fix $a \in \mathcal{A}$ with finite positive weights, and fix a finite $\lambda_\eta \geq 0$. Then $F_t^{(a)}$ on $\mathbb{R}^{K \times d}$ is strictly convex and coercive and therefore has a unique unconstrained minimizer E_t^* . If E_t^* lies strictly inside the concentration bound, it is also the unique solution of the bounded subproblem. Exact unconstrained proximal gradient with the exact majorization backtracking rule ($\varepsilon_{\text{maj}} = 0$) converges to E_t^* , and its objective gap is $O(u^{-1})$ after u inner iterations.*

The proof rests on the two bounds

$$\begin{aligned}
\nabla^2 f_t(E) &= \text{blockdiag}_{k=1}^K \{N_k \text{Cov}_{\eta_k}(Y)\}, \\
0 \prec \nabla^2 f_t(E) &\preceq \text{blockdiag}_{k=1}^K \{N_k I_d\}, \\
f_{t,k}(\eta_k) &\geq (N_k - \|r_k\|_2) \|\eta_k\|_2 - \frac{N_k(d-1)}{2} \log \|\eta_k\|_2 + O(1) \longrightarrow \infty.
\end{aligned} \tag{22}$$

The $O(1)$ remainder in the last line is taken as $\|\eta_k\|_2 \rightarrow \infty$. Here blockdiag denotes block-diagonal assembly, Y denotes a unit-norm vMF random vector, and $f_{t,k}$ is the k th smooth block of f_t . The Hessian bound gives strict convexity and a Lipschitz gradient, while $\|r_k\|_2 < N_k$ makes the linear term in the last line dominate the logarithmic term. Standard proximal-gradient theory then gives uniqueness, convergence, and the stated objective rate. The complete argument is given in Appendix Appendix A.

The theorem concerns one fixed-responsibility subproblem. The reported code terminates the inner loop at a fixed relative-step tolerance or iteration limit, so a finite fitted path is an inexact numerical approximation to this limiting sequence. It describes the bounded implementation only when the cap is inactive along the inner sequence. An active cap would require a separate projected-proximal analysis (Beck and Teboulle, 2009; Parikh and Boyd, 2014).

For the current responsibilities, recall

$$Q_{\lambda_\eta}^{(a)}(\Theta \mid \tau^{(t)}) = Q(\Theta \mid \tau^{(t)}) - \lambda_\eta \mathcal{P}_a(E).$$

Proposition 6 (Penalized GEM monotonicity). *Let $\tau^{(t)}$ be the posterior responsibilities under $\Theta^{(t)}$. If a candidate $\Theta^{(t+1)}$ satisfies*

$$Q_{\lambda_\eta}^{(a)}(\Theta^{(t+1)} \mid \tau^{(t)}) \geq Q_{\lambda_\eta}^{(a)}(\Theta^{(t)} \mid \tau^{(t)}),$$

then, in exact arithmetic,

$$\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t+1)}) \geq \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)}).$$

The implementation verifies both inequalities directly and permits decreases of at most 10^{-8} , so its finite trace is only numerically near-monotone. For the exact-arithmetic algorithm with zero acceptance tolerance, a bounded penalized criterion has a nondecreasing and therefore convergent sequence of

objective values. This monotonicity result alone does not establish boundedness or convergence of the full parameter sequence, stationary-point convergence, or global optimality. Theorem 2 gives a separate conditional stationarity result under explicit residual assumptions.

For a responsibility array τ , let f_τ denote the smooth fixed-responsibility objective in Section 3.2.1. Define

$$\begin{aligned}\mathcal{G}_s(E; \tau) &= \frac{1}{s} \left[E - \text{prox}_{s\lambda_\eta \mathcal{P}_a} \{E - s\nabla f_\tau(E)\} \right], \\ \mathcal{R}_\pi(\pi; \tau) &= \pi - \frac{N(\tau)}{n}, \quad N_k(\tau) = \sum_{i=1}^n \tau_{ik}.\end{aligned}\tag{23}$$

For Theorem 2, s_t is the diagnostic step size used to evaluate the natural-parameter residual at the final accepted inner iterate; it need not equal every trial step used during backtracking.

Theorem 2 (Stationarity under vanishing block residuals). *Fix $a \in \mathcal{A}$ with finite positive weights and fix λ_η . Consider an idealized GEM sequence $\{\Theta^{(t)}\}$ for which*

$$\begin{aligned}\tau^{(t)} &= \tau(\Theta^{(t)}), \quad \Theta^{(t)}, \Theta^{(t+1)} \in \mathcal{C} \Subset \text{ri}(\mathcal{T}_{\kappa_{\max}}), \\ \inf_{t,k} N_k(\tau^{(t)}) &> 0, \quad 0 < \underline{s} \leq s_t \leq \bar{s} < \infty, \\ \|\Theta^{(t+1)} - \Theta^{(t)}\| &\longrightarrow 0,\end{aligned}\tag{24}$$

where ri denotes relative interior and $\mathcal{C} \Subset \text{ri}(\mathcal{T}_{\kappa_{\max}})$ means that $\mathcal{C}$ is a compact subset of that relative interior. Suppose further that the block residuals satisfy

$$\|\mathcal{G}_{s_t}(E^{(t+1)}; \tau^{(t)})\|_F + \|\mathcal{R}_\pi(\pi^{(t+1)}; \tau^{(t)})\|_2 \longrightarrow 0.\tag{25}$$

Then every accumulation point $\bar{\Theta} = (\bar{\pi}, \bar{E})$ satisfies the interior KKT conditions

$$\begin{aligned}0 &\in -\nabla_E \ell(\bar{\Theta}) + \lambda_\eta \partial \mathcal{P}_a(\bar{E}), \\ \nabla_\pi \ell(\bar{\Theta}) &= \nu \mathbf{1}_K, \quad \mathbf{1}_K^\top \bar{\pi} = 1.\end{aligned}\tag{26}$$

Thus every accumulation point is stationary for the finite-parameter penalized observed objective. The stationarity statement is equivariant under component-label permutation.

For any convergent subsequence, the assumptions imply, after taking a further subsequence if needed,

$$\begin{aligned} (\Theta^{(t_j+1)}, \tau^{(t_j)}, s_{t_j}) &\longrightarrow (\bar{\Theta}, \tau(\bar{\Theta}), \bar{s}), \\ \bar{E} &= \text{prox}_{\bar{s}\lambda_\eta \mathcal{P}_a} \{\bar{E} - \bar{s} \nabla f_{\tau(\bar{\Theta})}(\bar{E})\}. \end{aligned} \tag{27}$$

Equation (27) and the proximal fixed-point identity give the first condition in (26); Fisher’s identity for the observed-data score and the vanishing mixing residual give the remaining simplex conditions. The complete subsequence argument is in Appendix Appendix A.

This theorem is a conditional local result under (24)–(25), not a convergence claim for every finite-tolerance path. Fixed stopping tolerances, failed path points, active numerical bounds, absence of asymptotic regularity, or component degeneracy fall outside its assumptions; the empirical diagnostics for those cases are reported separately (Dempster et al., 1977; Green, 1990; Wu, 1983; Tseng, 2004).

3.6. Computational complexity

For dense X , one E-step costs $O(nKd)$; for a sparse matrix it costs $O\{K \text{nnz}(X)\}$, where $\text{nnz}(X)$ is the number of nonzero entries of X . Forming the sufficient statistics has the same order, and one centered proximal-gradient update costs $O(Kd)$. Excluding the input matrix, the main working memory is $O(nK + Kd)$, in addition to $O(nd)$ for dense X or $O\{\text{nnz}(X)\}$ for sparse X . If the single path evaluates L tuning points, T_{GEM} outer iterations, and T_{PG} inner updates on average, its leading cost is

$$O [LT_{\text{GEM}} \{K \text{nnz}(X) + T_{\text{PG}}Kd\}]$$

for sparse input, plus the costs of distinct-support refits. These are operation counts rather than runtime superiority claims. Both E-CGL and E-ACGL use $L = 240$ before early stopping. Multiple dense starts add their own E-step and M-step costs before the path is constructed.

4. Numerical studies

4.1. Design and evaluation criteria

The primary experiment evaluates recovery of posterior-score contrast support under varying overlap and concentration heterogeneity. Observations

are drawn from a balanced $K = 4$ vMF mixture in $d = 200$ dimensions. The coordinates are partitioned as

$$\{1, \dots, d\} = S_C \cup S_D \cup S_N, \quad (q_C, q_D, q_N) = (4, 16, 180),$$

where S_C contains common natural-parameter coefficients, S_D contains centered component contrasts, and S_N is zero. Let $q_{\text{block}} = 4$ be the number of decision coordinates assigned to each component block. For a decision coordinate j in block g , define

$$u_{kj} = \begin{cases} 1, & k = g, \\ -1/(K - 1), & k \neq g. \end{cases}$$

For a candidate common-background norm A , the natural parameters are generated by

$$\eta_{kj}(A) = \begin{cases} A/\sqrt{q_C}, & j \in S_C, \\ s_k(A)u_{kj}, & j \in S_D, \\ 0, & j \in S_N, \end{cases} \quad s_k(A) = \frac{\sqrt{\kappa_k^2 - A^2}}{\|u_{k\cdot}\|_2}.$$

Hence $\|\eta_k(A)\|_2 = \kappa_k$ and $\mu_k = \eta_k/\kappa_k$. The clustering difficulty is calibrated by the oracle Bayes error

$$e_B = \Pr_{\Theta^*} \left\{ \arg \max_k \Pr(Z = k \mid X; \Theta^*) \neq Z \right\}.$$

We use $e_B \in \{0.025, 0.05, 0.10\}$, $n \in \{300, 1000\}$, and either

$$\kappa = (45, 45, 45, 45) \quad \text{or} \quad \kappa = (30, 40, 50, 60).$$

The common-to-contrast allocation is calibrated by Monte Carlo bisection while the prescribed concentrations and support sets are held fixed. Independent validation samples of size 200,000 gave achieved errors between 0.02566 and 0.10035; the largest absolute discrepancy from its target was 0.00172.

Each of the 12 cells contains 100 paired repetitions and an independent test sample of size 5,000. The common comparison set consists of spherical k -means, dense vMF mixtures with shared and component-specific concentrations, the Rossi-style entrywise mean-direction lasso (M-L), and E-CGL.

Table 1: Methods, support targets, and roles in the numerical comparisons.

Method	Support target	Role
Spherical k -means	none	external clustering baseline
Dense vMF, shared κ	none	concentration-restricted baseline
Dense vMF, free κ_k	none	likelihood baseline
M-L	S_P	published prototype-sparsity reference
M-CGL	S_μ	directional companion
E-CGL	S_η	primary proposed estimator
E-ACGL	S_η	secondary adaptive extension

The centered mean-direction group lasso (M-CGL) is included as a directional companion in the four prespecified $e_B = 0.05$ cells. E-CGL and E-ACGL use the single 240-point, zero-augmented KKT-scaled path in (16). M-CGL uses a separate 240-point dense-score-proxy path from $10^{-4}\lambda_{\text{proxy},\mu}$ to $1.25\lambda_{\text{proxy},\mu}$, as defined in Supplementary Section S8. M-L retains its Rossi-style event-driven path, with at most 240 updates in the simulation, because its threshold events are not indexed by either centered-group path scale. All mixture fits use 10 dense starts and Rcpp helpers; sparse mixture fits use BIC after target-specific support refitting. E-ACGL was evaluated in all 12 cells as a secondary adaptive extension.

Table 1 summarizes the estimand and role of each method. The common comparison set is evaluated in all 12 primary cells; M-CGL is restricted to the four prespecified $e_B = 0.05$ companion cells.

For clarity, the two mean-direction comparators use different penalties. The Rossi-style entrywise prototype penalty is

$$P_{\text{M-L}}(M) = \lambda_\mu \sum_{k=1}^K \sum_{j=1}^d |\mu_{kj}|, \quad \|\mu_k\|_2 = 1,$$

whereas the directional companion uses

$$P_{\text{M-CGL}}(M) = \lambda_\mu \sum_{j=1}^d \|H_K M_{\cdot j}\|_2, \quad \|\mu_k\|_2 = 1.$$

The primary simulations estimate component-specific concentrations for M-L. The Rossi-style bridge and the real-data M-L analysis use a shared concentration, as prespecified for those comparisons. M-CGL is fitted by introducing $\Xi = H_K M$ and alternating a product-of-spheres direction update,

centered group thresholding, and a scaled-dual update before numerical concentration updates. Its target-preserving refit enforces $H_K M_{\cdot j} = 0$ outside the selected directional-contrast support. The augmented-Lagrangian updates and numerical residual criteria are given in Supplementary Section S8.

Clustering is evaluated by the adjusted Rand index (ARI) and normalized mutual information (NMI). For the true partition $\mathcal{U}$ and fitted partition $\mathcal{V}$, the implementation uses

$$\text{NMI}(\mathcal{U}, \mathcal{V}) = \frac{I(\mathcal{U}; \mathcal{V})}{\{H(\mathcal{U})H(\mathcal{V})\}^{1/2}},$$

with value one when the two partitions are identical and both empirical entropies vanish, and zero when exactly one empirical entropy vanishes. Predictive fit is evaluated by the per-observation test negative log-likelihood (NLL)

$$\text{NLL}_{\text{test}} = -\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \log f(X_i; \hat{\Theta}),$$

where f is the vMF-mixture density with respect to surface-area measure on the unit sphere. Such a density can exceed one, so its reported NLL can be negative. Posterior-score contrast-support recovery is summarized first by the selected-coordinate counts

$$\hat{q}_C = |\hat{S}_\eta \cap S_C|, \quad \hat{q}_D = |\hat{S}_\eta \cap S_D|, \quad \hat{q}_N = |\hat{S}_\eta \cap S_N|,$$

and then by the true-positive rate (TPR), false-positive rate (FPR), precision, and F_1 score:

$$\text{TPR}_\eta = \frac{\hat{q}_D}{q_D}, \quad \text{FPR}_\eta = \frac{\hat{q}_C + \hat{q}_N}{q_C + q_N}, \quad \text{Precision}_\eta = \frac{\hat{q}_D}{|\hat{S}_\eta|},$$

and

$$F_{1,\eta} = \frac{2 \text{TPR}_\eta \text{Precision}_\eta}{\text{TPR}_\eta + \text{Precision}_\eta}, \quad \mathbf{1}\{\hat{S}_\eta = S_\eta^*\}.$$

Precision is set to zero when the estimated support is empty. If the true support is nonempty and the estimate is empty, TPR, precision, and F_1 are zero. More generally, F_1 is set to zero whenever $\text{TPR}_\eta + \text{Precision}_\eta = 0$. For an empty true support, these quantities are undefined and exact-empty recovery is reported instead. The exact-support rate is the Monte Carlo mean of the displayed indicator. Writing $\eta_{kj}^c = c_{kj} = \eta_{kj} - K^{-1} \sum_h \eta_{hj}$,

centered natural-parameter estimation is summarized by the mean squared error (MSE)

$$\text{MSE}_{\eta^c} = \frac{1}{Kd} \sum_{k=1}^K \sum_{j=1}^d (\hat{c}_{kj} - c_{kj}^*)^2,$$

after label alignment. Selected table entries are accompanied by Monte Carlo standard errors (MCSEs), computed across the 100 repetitions in each cell. The primary support tables evaluate every sparse procedure against S_η to answer a common posterior-score contrast question. Those entries are therefore cross-target diagnostics for M-L and M-CGL. Target-specific recovery is reported separately: M-L is evaluated against S_P in the summary below, and M-CGL is evaluated against S_μ in Section 4.3. Consequently, $F_{1,P}$, $F_{1,\mu}$, and $F_{1,\eta}$ are not interchangeable method rankings.

4.2. Posterior-score contrast support recovery

Tables 2 and 3 summarize every prespecified combination of oracle Bayes error, concentration pattern, and method, with estimation, posterior-score selection, and clustering quantities shown in separate column blocks. The tables include E-ACGL as a secondary adaptive row in every cell. Fig. 1 compares the replication-level ARI distributions, and Fig. 2 compares the corresponding $\log_{10}(\text{MSE}_{\eta^c})$ distributions for five likelihood-based comparators over the complete 12-cell grid, including E-ACGL as the secondary adaptive extension. In these common-target comparisons, support metrics are evaluated against S_η . M-L’s prototype-union recovery results and M-CGL’s directional-target results are interpreted separately as described above.

Across 11 of the 12 primary cells, E-CGL selected between 16.02 and 16.67 coordinates and attained $F_{1,\eta} \geq 0.981$. The exception combined $n = 300$, $e_B = 0.10$, and heterogeneous concentrations. In that cell E-CGL selected 20.97 coordinates and attained $F_{1,\eta} = 0.816$. At $n = 1000$, E-CGL had $F_{1,\eta} \geq 0.988$ in all six concentration-overlap cells. M-L retained 199.42–199.88 of 200 coordinates. Its cross-target $F_{1,\eta}$ was near 0.148. Against the 20-coordinate prototype-union target, M-L had $\text{TPR}_P = 1$, $\text{FPR}_P = 0.997$ – 0.999 , and $F_{1,P} = 0.182$ across the 12 cells, with no exact-support recovery. Thus, the low $F_{1,\eta}$ partly reflects a different estimand, while its prototype-union precision was also limited by near-dense selection. ARI decreased as e_B increased even when support recovery remained accurate, reflecting the calibrated increase in population overlap. Supplementary Tables S1–S2 add the complete common-target support decomposition, NMI, exact-

recovery rates, and failure summaries. The companion machine-readable file `supplementary_simulation_summary.csv` provides Monte Carlo standard errors and the M-L prototype-union metrics quoted above. Under the common single-path rule, E-ACGL selected 15.23–16.33 coordinates at $n = 300$ and 16.01–16.06 at $n = 1000$; its corresponding $F_{1,\eta}$ ranges were 0.947–0.996 and 0.998–1.000. These secondary results do not establish uniform improvement over E-CGL.

4.3. Directional versus posterior-score estimands

Four diagnostics separate directional and posterior-score heterogeneity. Under common concentration, $S_\mu = S_\eta$. Under pure concentration heterogeneity, the component directions are common, so $S_\mu = \emptyset$ while S_η contains 16 coordinates; this design uses $\kappa = (10, 30, 80, 200)$. The shared natural-parameter design uses $\kappa = (30, 40, 50, 60)$ and contains 80 common natural-parameter coordinates and 20 posterior-score contrast coordinates. The crossed design uses the same heterogeneous concentrations and contains four η -only, four μ -only, eight jointly active, and eight null coordinates. Table 4 compares recovery against both population targets in these four designs.

For the empty directional target, support F_1 is undefined and “Exact support” is the probability of selecting the empty support. M-CGL selected that support in 20% of the pure- concentration repetitions, whereas E-CGL recovered the nonempty posterior-score contrast support with mean $F_{1,\eta} = 0.992$. In the crossed design, each method had high target-specific F_1 and lower F_1 against the other target. The shared natural-parameter design exposed a boundary for the directional companion: its exact support was not selected in any replication. These results motivate target-specific, rather than winner-takes-all, comparisons.

4.4. Sample-size recovery and oracle benchmarks

Table 5 gives the complete numerical trajectories, and Fig. 3 displays their trends. E-CGL is followed over $n \in \{300, 600, 1000, 2000\}$. Both $e_B = 0.05$ and $e_B = 0.10$ include equal and heterogeneous concentration trajectories. At $e_B = 0.05$, centered- η MSE decreased from 0.220 to 0.028 under equal concentrations and from 0.232 to 0.028 under heterogeneous concentrations. FPR decreased to zero and exact-support recovery reached one at $n = 2000$. Under $e_B = 0.10$ and equal concentrations, E-CGL moved from $\hat{q} = 16.28$, $F_{1,\eta} = 0.992$, and exact recovery 0.77 at $n = 300$ to $\hat{q} = 16.00$, $F_{1,\eta} = 1.000$, and exact recovery 1.00 at $n = 2000$; MSE_{η^c} decreased from 0.253 to 0.031.

Table 2: Primary method comparison at $n = 300$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

e_B	κ pattern	method	Estimation	Posterior-score selection				Clustering and fit	
			MSE $_{\eta^c}$ (100 $\times$ MCSE)	$\hat{q}$	FPR $_{\eta}$	TPR $_{\eta}$	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k -means	—	—	—	—	—	0.863	—
		Dense-S	2.471 (1.652)	—	—	—	—	0.869	-245.921
		Dense-F	2.505 (1.675)	—	—	—	—	0.864	-245.906
		M-L	2.447 (1.675)	199.58	0.998	1.000	0.148	0.865	-245.934
		E-CGL	0.198 (0.402)	16.14	0.001	1.000	0.996	0.927	-246.935
		E-ACGL	0.198 (0.402)	16.14	0.001	1.000	0.996	0.927	-246.935
0.025	heterogeneous	Sph. k -means	—	—	—	—	—	0.775	—
		Dense-S	3.092 (1.986)	—	—	—	—	0.798	-245.869
		Dense-F	2.540 (1.702)	—	—	—	—	0.856	-246.110
		M-L	2.475 (1.648)	199.61	0.998	1.000	0.148	0.860	-246.138
		E-CGL	0.192 (0.512)	16.15	0.001	1.000	0.996	0.925	-247.153
		E-ACGL	0.192 (0.512)	16.15	0.001	1.000	0.996	0.925	-247.153
0.050	equal	Sph. k -means	—	—	—	—	—	0.620	—
		Dense-S	2.810 (2.275)	—	—	—	—	0.738	-245.936
		Dense-F	2.863 (2.347)	—	—	—	—	0.726	-245.919
		M-L	2.771 (2.133)	199.65	0.998	1.000	0.148	0.731	-245.954
		M-CGL	0.222 (0.585)	16.24	0.001	1.000	0.993	0.856	-247.009
		E-CGL	0.220 (0.599)	16.25	0.001	1.000	0.992	0.856	-247.010
0.050	heterogeneous	E-ACGL	0.219 (0.577)	16.23	0.001	1.000	0.993	0.856	-247.011
		Sph. k -means	—	—	—	—	—	0.580	—
		Dense-S	3.764 (4.549)	—	—	—	—	0.649	-245.860
		Dense-F	3.077 (4.439)	—	—	—	—	0.709	-246.082
		M-L	2.957 (4.432)	199.70	0.998	1.000	0.148	0.718	-246.122
		M-CGL	0.615 (2.590)	20.90	0.034	0.917	0.797	0.815	-247.064
0.100	equal	E-CGL	0.232 (0.822)	16.67	0.004	1.000	0.981	0.857	-247.205
		E-ACGL	0.209 (0.555)	16.18	0.001	1.000	0.995	0.858	-247.215
		Sph. k -means	—	—	—	—	—	0.211	—
		Dense-S	4.191 (9.094)	—	—	—	—	0.436	-245.788
		Dense-F	4.509 (10.058)	—	—	—	—	0.390	-245.737
		M-L	4.143 (9.850)	199.67	0.998	1.000	0.148	0.422	-245.830
0.100	heterogeneous	E-CGL	0.253 (0.660)	16.28	0.002	1.000	0.992	0.724	-247.130
		E-ACGL	0.273 (1.544)	16.33	0.002	0.997	0.987	0.721	-247.124
		Sph. k -means	—	—	—	—	—	0.444	—
		Dense-S	4.768 (4.502)	—	—	—	—	0.484	-245.998
		Dense-F	4.591 (7.985)	—	—	—	—	0.512	-246.148
		M-L	4.529 (8.296)	199.62	0.998	1.000	0.148	0.514	-246.173
		E-CGL	0.871 (3.751)	20.97	0.033	0.926	0.816	0.636	-247.164
		E-ACGL	0.419 (2.072)	15.23	0.002	0.925	0.947	0.702	-247.283

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE $_{\eta^c}$ multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_{η} for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 3: Primary method comparison at $n = 1000$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

e_B	κ pattern	method	Estimation	Posterior-score selection				Clustering and fit	
			MSE $_{\eta^c}$ (100 $\times$ MCSE)	$\hat{q}$	FPR $_{\eta}$	TPR $_{\eta}$	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k -means	—	—	—	—	—	0.917	—
		Dense-S	0.676 (0.347)	—	—	—	—	0.918	-246.966
		Dense-F	0.679 (0.342)	—	—	—	—	0.917	-246.964
		M-L	0.672 (0.339)	199.42	0.997	1.000	0.149	0.918	-246.968
		E-CGL	0.055 (0.110)	16.02	0.000	1.000	0.999	0.933	-247.245
		E-ACGL	0.055 (0.110)	16.02	0.000	1.000	0.999	0.933	-247.245
0.025	heterogeneous	Sph. k -means	—	—	—	—	—	0.866	—
		Dense-S	1.147 (0.534)	—	—	—	—	0.861	-246.931
		Dense-F	0.685 (0.404)	—	—	—	—	0.912	-247.167
		M-L	0.677 (0.405)	199.57	0.998	1.000	0.148	0.913	-247.170
		E-CGL	0.056 (0.115)	16.02	0.000	1.000	0.999	0.928	-247.451
		E-ACGL	0.056 (0.115)	16.02	0.000	1.000	0.999	0.928	-247.451
0.050	equal	Sph. k -means	—	—	—	—	—	0.835	—
		Dense-S	0.731 (0.436)	—	—	—	—	0.837	-247.035
		Dense-F	0.737 (0.437)	—	—	—	—	0.836	-247.032
		M-L	0.726 (0.437)	199.58	0.998	1.000	0.148	0.837	-247.036
		M-CGL	0.061 (0.150)	16.04	0.000	1.000	0.999	0.867	-247.316
		E-CGL	0.061 (0.161)	16.06	0.000	1.000	0.998	0.867	-247.316
0.050	heterogeneous	Sph. k -means	—	—	—	—	—	0.773	—
		Dense-S	1.273 (0.642)	—	—	—	—	0.765	-247.004
		Dense-F	0.747 (0.476)	—	—	—	—	0.835	-247.228
		M-L	0.736 (0.478)	199.67	0.998	1.000	0.148	0.836	-247.232
		M-CGL	0.087 (0.143)	20.04	0.022	1.000	0.888	0.867	-247.510
		E-CGL	0.061 (0.132)	16.02	0.000	1.000	0.999	0.869	-247.517
0.100	equal	Sph. k -means	—	—	—	—	—	0.676	—
		Dense-S	0.858 (0.482)	—	—	—	—	0.684	-247.146
		Dense-F	0.866 (0.490)	—	—	—	—	0.682	-247.144
		M-L	0.851 (0.492)	199.64	0.998	1.000	0.148	0.683	-247.149
		E-CGL	0.067 (0.133)	16.02	0.000	1.000	0.999	0.747	-247.445
		E-ACGL	0.067 (0.133)	16.02	0.000	1.000	0.999	0.747	-247.445
0.100	heterogeneous	Sph. k -means	—	—	—	—	—	0.589	—
		Dense-S	1.758 (1.744)	—	—	—	—	0.595	-247.110
		Dense-F	1.013 (1.822)	—	—	—	—	0.669	-247.308
		M-L	0.970 (1.583)	199.88	0.999	1.000	0.148	0.675	-247.316
		E-CGL	0.077 (0.198)	16.41	0.002	1.000	0.988	0.753	-247.627
		E-ACGL	0.071 (0.155)	16.01	0.000	1.000	1.000	0.753	-247.630

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE $_{\eta^c}$ multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_{η} for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 4: Directional and posterior-score contrast recovery in the estimand-separation designs. Entries are means over 100 replications, and the two target-specific F_1 scores are evaluated against their respective population targets. The exact-support column uses S_μ for M-CGL and S_η for E-CGL.

Design	Method	q_μ^*	q_η^*	$\hat{q}$	$F_{1,\mu}$	$F_{1,\eta}$	Exact support	ARI
common κ	M-CGL	16	16	16.04	0.999	0.999	0.96	0.867
common κ	E-CGL	16	16	16.06	0.998	0.998	0.95	0.867
pure concentration	M-CGL	0	16	0.96	–	0.005	0.20	0.675
pure concentration	E-CGL	0	16	16.27	–	0.992	0.83	0.633
shared natural parameter	M-CGL	100	20	21.13	0.348	0.974	0.00	0.859
shared natural parameter	E-CGL	100	20	20.02	0.333	1.000	0.98	0.870
crossed support	M-CGL	12	12	11.68	0.984	0.677	0.64	0.996
crossed support	E-CGL	12	12	11.89	0.681	0.980	0.69	0.996

Under the heterogeneous counterpart, E-CGL moved from $\hat{q} = 20.97$, $F_{1,\eta} = 0.816$, and exact recovery 0.02 at $n = 300$ to $\hat{q} = 16.03$, $F_{1,\eta} = 0.999$, and exact recovery 0.97 at $n = 2000$.

The path oracle is the fitted candidate with the largest $F_{1,\eta}$ within each repetition. The oracle- S_η estimator refits under the known population support. We report

$$\Delta_{\text{sel}} = F_1^{\text{path oracle}} - F_1^{\text{BIC}}, \quad \Delta_{\text{NLL}} = \text{NLL}(\hat{\Theta}) - \text{NLL}(\hat{\Theta}_{S_\eta}^{\text{oracle support}}).$$

These finite-sample benchmark gaps are not claims about asymptotic support recovery. At $e_B = 0.05$, all E-CGL selector gaps were at most 0.0083 at $n = 300$ and at most 0.0018 for $n \geq 600$. The difficult heterogeneous- κ , $e_B = 0.10$ trajectory had $\Delta_{\text{sel}} = 0.0478$ and $\Delta_{\text{NLL}} = 0.1653$ at $n = 300$. The NLL gap decreased with sample size; the selector gap rose slightly at $n = 600$ to 0.0502 before decreasing to 0.0009 at $n = 2000$. The BIC, path-oracle, and oracle-support-refit comparisons are reported in Supplementary Table S4.

4.5. Stress conditions and computation

Five stress cells examine moderately dense support at $n = 300$ and $n = 1000$, strongly dense support, $d > n$, and a weak beta-min condition. In the sparse weak-signal cell, E-CGL retained mean $F_{1,\eta} = 0.998$ and exact recovery 0.93. In the $d = 500, n = 300$ cell, E-CGL’s $F_{1,\eta}$ was 0.803. For 80 active coordinates at $n = 300$, E-CGL selected 81.98 coordinates on average, with $F_{1,\eta} = 0.805$ and ARI 0.556. At $n = 1000$, it selected 80.42 coordinates and attained $F_{1,\eta} = 0.900$. None of the final E-series fits selected the

Table 5: E-CGL sample-size recovery over 100 repetitions. Parenthesized values are Monte Carlo standard errors multiplied by 100.

Panel A: centered natural-parameter contrast estimation

e_B	κ pattern	n	$\hat{q}$	MSE_{η^c} ($100 \times \text{MCSE}$)	$n \text{ MSE}_{\eta^c}$
0.05	equal	300	16.25	0.220 (0.599)	66.14
		600	16.02	0.097 (0.215)	58.24
		1000	16.06	0.061 (0.161)	61.26
		2000	16.00	0.028 (0.055)	56.45
0.05	heterogeneous	300	16.67	0.232 (0.822)	69.47
		600	16.06	0.099 (0.245)	59.68
		1000	16.02	0.061 (0.132)	61.40
		2000	16.00	0.028 (0.061)	55.64
0.10	equal	300	16.28	0.253 (0.660)	75.78
		600	16.03	0.110 (0.243)	66.18
		1000	16.02	0.067 (0.133)	66.56
		2000	16.00	0.031 (0.063)	62.76
0.10	heterogeneous	300	20.97	0.871 (3.751)	261.23
		600	18.20	0.194 (0.991)	116.42
		1000	16.41	0.077 (0.198)	77.10
		2000	16.03	0.035 (0.079)	69.96

Panel B: posterior-score support and clustering

e_B	κ pattern	n	TPR_{η} ($100 \times \text{MCSE}$)	FPR_{η} ($100 \times \text{MCSE}$)	$F_{1,\eta}$	exact support	ARI
0.05	equal	300	1.000 (0.000)	0.001 (0.028)	0.992	0.78	0.856
		600	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.864
		1000	1.000 (0.000)	0.000 (0.015)	0.998	0.95	0.867
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.869
0.05	heterogeneous	300	1.000 (0.000)	0.004 (0.072)	0.981	0.67	0.857
		600	1.000 (0.000)	0.000 (0.015)	0.998	0.95	0.866
		1000	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.869
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.871
0.10	equal	300	1.000 (0.000)	0.002 (0.030)	0.992	0.77	0.724
		600	1.000 (0.000)	0.000 (0.009)	0.999	0.97	0.742
		1000	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.747
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.749
0.10	heterogeneous	300	0.926 (0.773)	0.033 (0.344)	0.816	0.02	0.636
		600	0.993 (0.250)	0.013 (0.141)	0.934	0.28	0.739
		1000	1.000 (0.000)	0.002 (0.047)	0.988	0.74	0.753
		2000	1.000 (0.000)	0.000 (0.009)	0.999	0.97	0.757

Notes: Point estimates are reported on their original scales. Parenthesized MCSE values for MSE_{η^c} , TPR_{η} , and FPR_{η} are multiplied by 100 and reported to three decimal places. The true posterior-score support size is 16. Both $e_B = 0.05$ and $e_B = 0.10$ include equal and heterogeneous concentration trajectories.

explicit empty support in the five stress cells. With 160 active coordinates, E-CGL selected 151.92 on average and had $F_{1,\eta} = 0.921$. E-ACGL reduced false-positive selection in several stress cells but did not uniformly improve estimation or clustering. Full stress results and method-phase convergence summaries are reported in Table 6; Supplementary Tables S5–S6 add test NLL, Monte Carlo standard errors, convergence, path-endpoint, and timing diagnostics.

In a paired runtime diagnostic for the representative heterogeneous $e_B = 0.05$, $n = 1000$, $d = 200$ cell, median end-to-end wall-clock times and interquartile ranges (IQRs) over 100 repetitions were 509.61 (415.62–597.18) seconds for M-CGL and 74.62 (69.14–79.85) seconds for E-CGL. The sparse-method totals include the paired dense initialization and the method-specific path, refit, and selector; one-time Rcpp compilation was excluded. Because the methods use different path and optimization rules, these values describe computational burden rather than establishing speed superiority. The full seven-method timing panel under the final method-specific path rules is reported in Supplementary Table S15.

Convergence, failure, path-endpoint, and method-phase timing diagnostics are reported in Supplementary Tables S6 and S15. These checks are numerical diagnostics and do not establish global optimality.

Table 6: Stress-condition results over 100 repetitions. Parenthesized scenario triplets give (n, d, q^*) .

Scenario	method	$\hat{q}$	TPR_η	FPR_η	$F_{1,\eta}$	exact	ARI	MSE_{η^c}
Moderately dense (300, 200, 80)	Sph. k -means	—	—	—	—	—	0.449	—
	Dense-S	—	—	—	—	—	0.481	4.830
	Dense-F	—	—	—	—	—	0.512	4.393
	M-L	199.50	1.000	0.996	0.572	0.00	0.515	4.360
	E-CGL	81.98	0.814	0.141	0.805	0.00	0.556	5.009
	E-ACGL	58.03	0.688	0.025	0.797	0.00	0.544	3.409
Moderately dense (1000, 200, 80)	Sph. k -means	—	—	—	—	—	0.586	—
	Dense-S	—	—	—	—	—	0.600	1.719
	Dense-F	—	—	—	—	—	0.670	0.989
	M-L	199.81	1.000	0.998	0.572	0.00	0.673	0.957
	E-CGL	80.42	0.902	0.068	0.900	0.00	0.650	0.816
	E-ACGL	67.53	0.843	0.001	0.914	0.00	0.679	0.641
Strongly dense (1000, 200, 160)	Sph. k -means	—	—	—	—	—	0.589	—
	Dense-S	—	—	—	—	—	0.598	1.716
	Dense-F	—	—	—	—	—	0.669	0.998
	M-L	199.73	1.000	0.993	0.890	0.00	0.671	0.968
	E-CGL	151.92	0.898	0.204	0.921	0.00	0.629	1.985
	E-ACGL	125.56	0.783	0.007	0.877	0.00	0.612	5.687
High-dimensional (300, 500, 40)	Sph. k -means	—	—	—	—	—	0.285	—
	Dense-S	—	—	—	—	—	0.405	9.863
	Dense-F	—	—	—	—	—	0.424	9.519
	M-L	499.99	1.000	1.000	0.148	0.00	0.431	9.441
	E-CGL	54.55	0.926	0.038	0.803	0.00	0.759	1.580
	E-ACGL	36.74	0.853	0.006	0.888	0.00	0.766	1.252
Weak beta-min (1000, 200, 16)	Sph. k -means	—	—	—	—	—	0.776	—
	Dense-S	—	—	—	—	—	0.768	1.269
	Dense-F	—	—	—	—	—	0.837	0.750
	M-L	199.72	1.000	0.998	0.148	0.00	0.838	0.740
	E-CGL	15.99	0.998	0.000	0.998	0.93	0.869	0.059
	E-ACGL	15.99	0.998	0.000	0.998	0.93	0.869	0.059

Notes: Support metrics use S_η . Dashes indicate that a method does not return a coordinate support. The strongly dense cell is a negative control rather than a setting tailored to sparse selection. Monte Carlo standard errors, test NLL, convergence, endpoint, and runtime summaries are reported in the Supplement.

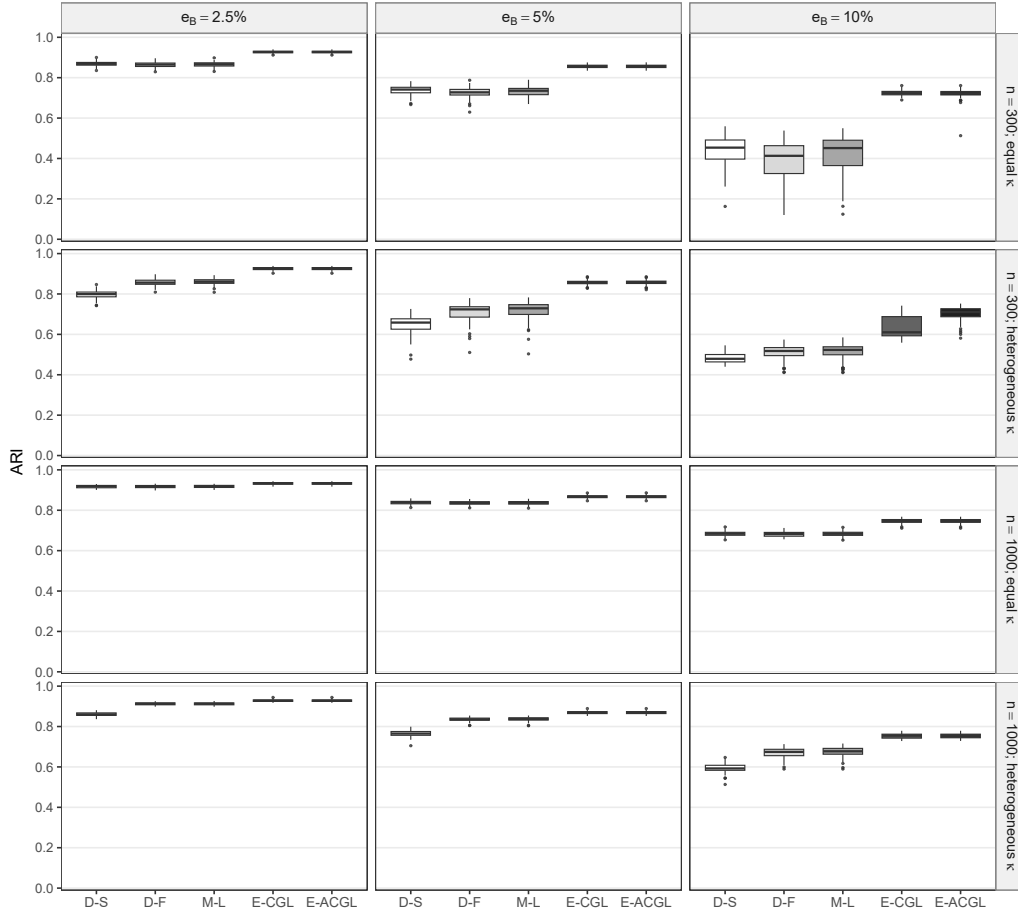


Figure 1: Clustering accuracy across the primary simulation grid. Boxplots show ARI over 100 paired replications in each cell. Columns give the target oracle Bayes error, and rows give sample size and concentration pattern. D-S and D-F denote dense vMF mixtures with shared and component-specific concentrations, respectively; E-ACGL is the secondary adaptive extension. M-CGL is omitted because it was evaluated only in the four prespecified $e_B = 0.05$ cells; its target-specific results are reported in Table 4. Spherical k -means is omitted from the figure for readability and is reported in Tables 2 and 3.

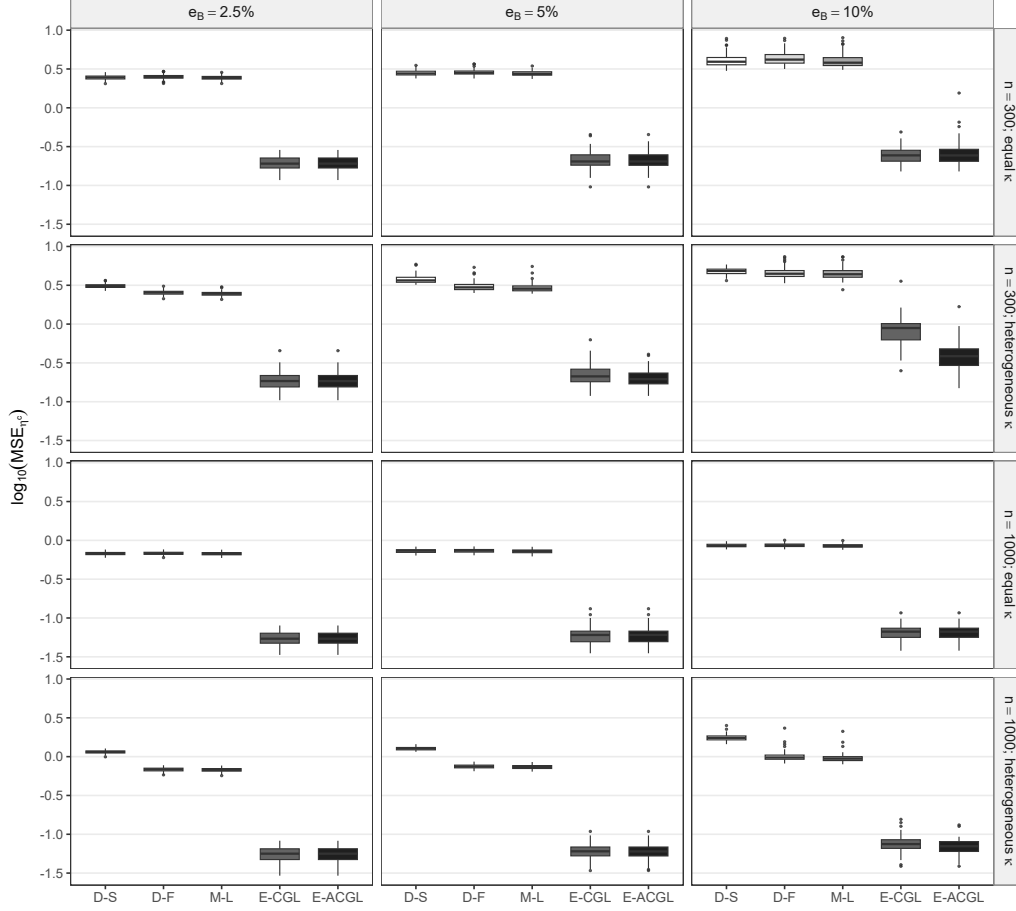


Figure 2: Centered natural-parameter estimation error across the primary simulation grid. Boxplots show $\log_{10}(\text{MSE}_{\eta^c})$ over 100 paired replications in each cell, where $\eta^c = H_K E$. Columns give the target oracle Bayes error, and rows give sample size and concentration pattern. D-S and D-F denote dense vMF mixtures with shared and component-specific concentrations, respectively. E-ACGL is the secondary adaptive extension. M-CGL is omitted because it was evaluated only in the four prespecified $e_B = 0.05$ cells; its target-specific results are reported in Table 4.

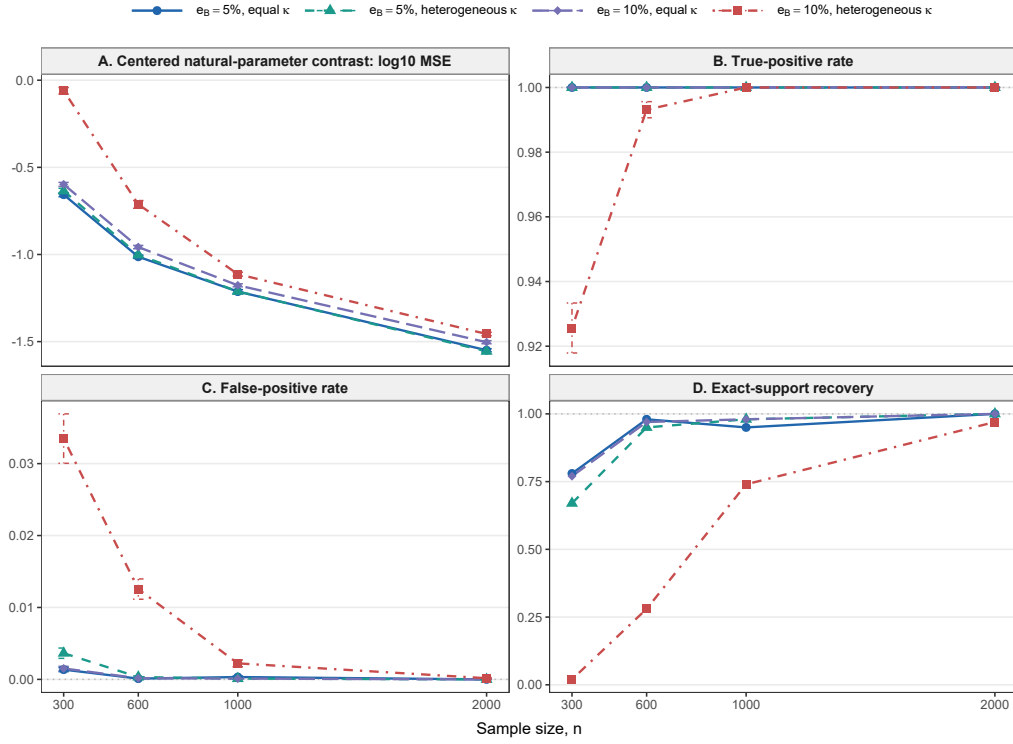


Figure 3: Sample-size trajectories for E-CGL. Points show Monte Carlo means, and error bars show Monte Carlo standard errors over 100 replications. Both oracle-error levels are shown under equal and heterogeneous concentrations.

5. Real data analysis

5.1. Data and analysis protocol

Classic3 contains abstracts from the CISI, CRAN, and MED document collections and has been used in directional document clustering (Banerjee et al., 2005). The raw payload contained $n_0 = 3,890$ documents. We removed two exact duplicates and five prespecified near-duplicates, leaving $n = 3,883$ documents. The cleaned class counts were 1,454, 1,397, and 1,032, respectively.

Documents were encoded with the vocabulary-aligned Sparse Lexical and Expansion Model (SPLADE), with a maximum input sequence length of 512 (Formal et al., 2021, 2022). The exact encoding identifiers were

```
Model: naver/splade-cocondenser-ensembledistil  
Revision: 49cf4c7b0db5b870a401ddf5e2669993ef3699c7.
```

The payload and vocabulary checksums are reported in Supplementary Section S9. Coordinates occurring in fewer than two documents were removed, the top 2,000 coordinates were ranked by variance over the unlabeled cleaned payload, and each vector was normalized to unit Euclidean norm. The resulting matrix had no zero rows, duplicate transformed rows, duplicate vocabulary entries, or nonfinite values. The model identifier, revision, vocabulary hash, duplicate audit, and payload hash were fixed before model fitting.

The preprocessing rules and split indices were prespecified. E-CGL uses the same single 240-point path and explicit empty-support candidate as in the primary simulations.

The benchmark resolution was fixed at $K = 3$ to match the three supplied broad topic labels. The labels were otherwise withheld from coordinate ranking, initialization, parameter estimation, tuning, and support selection; they were used after fitting only to calculate ARI and NMI and to name aligned components. Five prespecified 80/20 splits provide stability and held-out diagnostics; they are not pooled with the full-data estimates. After the prespecified duplicate exclusions, each split ranks coordinates using its training documents only. The resulting training-selected coordinate mask is then applied without change to the test documents, after which both matrices are row-normalized.

5.2. Methods and evaluation

The comparison comprised spherical k -means, permutation-tuned sparse k -means (Witten and Tibshirani, 2010), dense vMF mixtures with shared

and component-specific concentrations, the Rossi-style M-L prototype selector, M-CGL, E-CGL, and the secondary E-ACGL extension. Likelihood-based fits used 30 dense random starts. The M-L path allowed 600 Rossi-style threshold updates. E-CGL and E-ACGL used the single zero-augmented 240-point relative path defined in Section 3; M-CGL used the 240-point dense-score-proxy path defined in Supplementary Section S8. Each was followed by target-specific distinct-support refitting, an explicit empty-support candidate, and within-method BIC selection. E-ACGL used the fixed weights in (1) and is reported as a secondary analysis rather than used to redefine or rank the primary E-CGL estimator. The same E-series rule was used in each prespecified Classic3 split. The sparse vMF methods were selected by their method-specific BIC after support-constrained refitting. Sparse k -means used a label-blind permutation gap criterion with 25 permutations and 10 weight-bound candidates.

Selected q is target-specific: weighted coordinates for sparse k -means, prototype-union coordinates for M-L, directional-contrast coordinates for M-CGL, and posterior-score contrast coordinates for E-CGL and E-ACGL. Accordingly, differences in q do not by themselves rank the estimands. All methods were evaluated by ARI and NMI. Observed log-likelihood and the practical nominal-dimension BIC were additionally reported for vMF models.

5.3. Full-data results and interpretation

The dense free- κ_k mixture attained ARI 0.9760 and NMI 0.9538. E-CGL selected 1,420 of 2,000 coordinates, excluding 29.00% of the coordinates from posterior-score contrasts while attaining ARI 0.9761 and NMI 0.9539. M-CGL produced a closely matching partition (ARI 0.976 and NMI 0.954) with 1,471 directional-contrast coordinates. M-L retained 1,942 prototype coordinates. E-ACGL selected 1,438 centered natural-parameter coordinates and attained ARI 0.9746 and NMI 0.9515. Its result is reported as a secondary adaptive analysis; the small performance differences from E-CGL are not interpreted as evidence of superiority. The permutation-tuned sparse k -means fit selected 985 coordinates but had ARI 0.1374; its selected weight bound was not at the evaluated grid boundary. Independent restarts at that bound reproduced ARI 0.1373 and nearly the same objective value, so this result is reported as performance of the permutation-selected solution under the SPLADE representation rather than as a single-start anomaly or a general statement about sparse k -means. Table 7 summarizes the full-data comparison.

Table 7: Clustering and selected support in the full-data Classic3 analysis. The data contain $n = 3,883$ documents and $d = 2,000$ coordinates after duplicate removal; $K = 3$, and component order is aligned to CISI, CRAN, and MED only after fitting.

Method	q (q/d)	Cluster sizes	ARI	NMI	ℓ	BIC	$\hat{\kappa}$
Spherical k-means	2000 (1.000)	1458/1406/1019	0.970	0.944	—	—	—
Sparse k-means	985 (0.492)	553/784/2546	0.137	0.313	—	—	—
Dense vMF shared κ	2000 (1.000)	1458/1406/1019	0.970	0.944	18,917,075	-37,784,564	732.2/732.2/732.2
Dense vMF free κ_k	2000 (1.000)	1462/1391/1030	0.976	0.954	18,921,337	-37,793,072	738.5/818.0/615.3
M-L	1942 (0.971)	1462/1405/1016	0.970	0.944	18,914,871	-37,804,965	730.0/730.0/730.0
M-CGL	1471 (0.736)	1456/1392/1035	0.976	0.954	18,919,480	-37,798,101	738.9/816.5/610.0
E-CGL	1420 (0.710)	1456/1392/1035	0.976	0.954	18,919,114	-37,798,212	738.3/815.8/610.4
E-ACGL	1438 (0.719)	1458/1391/1034	0.975	0.951	18,919,276	-37,798,237	738.0/815.9/611.4

Selected q refers to the method-specific target: weighted coordinates for sparse k -means, prototype-union coordinates for M-L, directional-contrast coordinates for M-CGL, and posterior-score contrast coordinates for E-CGL and E-ACGL. E-CGL and E-ACGL use one zero-augmented $L = 240$ KKT-scaled path with relative lower endpoint 10^{-4} ; M-CGL uses its $L = 240$ dense-score-proxy path, and M-L retains its Rossi-style event-driven threshold path. Likelihood and BIC are not defined for the two k -means procedures. Nominal-dimension rules differ across methods, so cross-method BIC differences are descriptive; BIC is used primarily within each refitted candidate family. $\hat{\kappa}$ is ordered as CISI/CRAN/MED after post-fit component alignment. E-ACGL is a secondary adaptive analysis and is not used to rank the primary E-CGL estimator.

For coordinate j , define

$$R_j = \left(n^{-1} \sum_{i=1}^n x_{ij}^2 \right)^{1/2}, \quad I_{kj}^{(\text{ctr})} = R_j \left(\hat{\eta}_{kj} - \hat{\eta}_j \right), \quad I_j^{(\text{com})} = R_j \left| \hat{\eta}_j \right|.$$

The E-CGL concentrations, aligned to CISI, CRAN, and MED only after fitting, were 738.32, 815.85, and 610.42. Panel A of Fig. 4 selects the five largest positive root-mean-square (RMS)-scaled contributions for each matched component and displays their signed values $I_{kj}^{(\text{ctr})}$. CISI contrasts were led by *library*, *information*, and *librarian*; CRAN by *flow*, *mach*, and *pressure*; and MED by *tumor*, *inhibitor*, and *rat*. Panel B ranks inactive coordinates by $I_j^{(\text{com})}$ and shows those with the largest common root-mean-square-scaled natural-parameter contributions. Their fitted values are equal across components, so their centered contrasts are numerically zero even though the common natural-parameter baseline remains in the fitted density.

5.4. Classic3 split stability

For each pair of the five prespecified Classic3 splits, the conditional support Jaccard index is the support-intersection size divided by the support-union size after both supports are restricted to the vocabulary shared by that

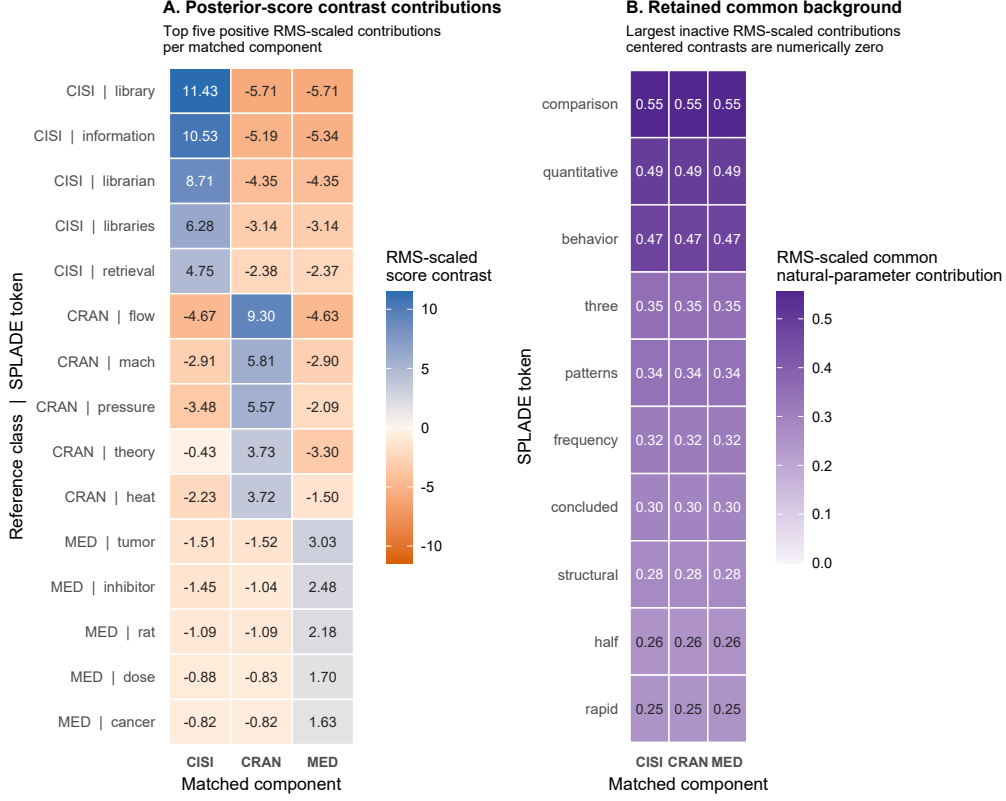


Figure 4: RMS-scaled posterior-score contrasts and common natural-parameter contributions in Classic3. E-CGL selected 1,420 of 2,000 coordinates. Panel A shows signed RMS-scaled posterior-score contrast contributions $I_{kj}^{(\text{ctr})} = R_j(\hat{\eta}_{kj} - \hat{\eta}_j)$ (blue, positive; orange, negative). Panel B shows the ten inactive coordinates having the largest $I_j^{(\text{com})} = R_j|\hat{\eta}_j|$. Their equal component values represent retained common effects rather than posterior-score heterogeneity. Supplied labels were used only after fitting to align and name the components.

pair. E-CGL attained a mean of 0.933 over the ten split pairs, and the 15 selected-contrast tokens displayed in Fig. 4 were retained in all five splits. Because the true coordinate support is unknown, support TPR, FPR, precision, and F_1 are not reported. Additional splitwise diagnostics and external-scope analyses are provided in the Supplementary Material.

6. Discussion

6.1. Posterior-score contrast support as the estimand

This study defines posterior-score contrast support as the coordinates entering the feature-dependent linear term of pairwise posterior log-score contrasts. E-CGL regularizes centered natural-parameter contrasts and therefore targets posterior-score contrast coordinates. M-L targets prototype support, whereas the M-CGL companion targets directional-contrast support. Centering leaves the common natural-parameter baseline unpenalized, so a coordinate shared across components is not selected solely because it remains nonzero in every prototype. The coordinate-group penalty is justified by alignment between the regularization unit and this estimand, rather than by uniform empirical superiority over entrywise regularization or prototype-oriented penalties.

6.2. Empirical scope

Across the evaluated $n = 1000$ primary-grid cells, E-CGL selected 16.02–16.41 coordinates on average and attained $F_{1,\eta}$ between 0.988 and 0.999 for a true support size of 16. Clustering ARI decreased with increasing oracle Bayes error, whereas support recovery changed little in these cells; clustering and support difficulty therefore need not vary in parallel in the evaluated settings. The shared natural-parameter experiment illustrated the practical difference between directional and posterior-score contrast support. The dense-support experiment, in contrast, documented substantial finite-sample under-selection when decision information was distributed across many weak coordinates, although recovery improved at $n = 1000$ in that design.

E-ACGL is included as a secondary adaptive extension. Its compact simulation and Classic3 results are reported in the main article. These analyses do not establish an adaptive-weighting advantage and do not redefine the primary E-CGL estimator. The primary motivation for E-CGL is alignment between the coordinate-group penalty and the posterior-score contrast support target.

The target-separation study in Section 4.3 evaluated M-CGL against directional-contrast support and E-CGL against posterior-score contrast support. Both methods recovered their common target under common concentration. In the pure-concentration design, E-CGL recovered the nonempty posterior-score contrast support with mean $F_{1,\eta} = 0.992$, whereas M-CGL

selected the correct empty directional-contrast support in 20 of 100 repetitions. In the shared natural-parameter design, E-CGL recovered the smaller posterior-score contrast target, whereas M-CGL underselected the larger directional target and did not recover it exactly on the recorded path. These results document path and numerical sensitivity for the directional companion and motivate its diagnostic rather than co-primary role.

The Classic3 analyses, with prespecified preprocessing and split indices, provided complementary evidence through full-data clustering, coordinate reduction, interpretation, and splitwise support stability rather than recovery of an unknown true support. In the full Classic3 fit, E-CGL excluded 29.00% of the coordinates while closely matching the partition produced by the dense component-specific-concentration model. Across the five prespecified splits, its conditional support Jaccard index was 0.933, and the 15 selected-contrast tokens shown in Fig. 4 were retained in all five splits. These results assess stability under the prespecified representation; they do not establish recovery of an unknown true support.

6.3. Model selection and computation

The support-recovery claims condition on a fixed number of components. The separate two-stage K diagnostic in Section 3.3.4 is a practical procedure and does not establish consistency of component-number selection. When density criteria and partition stability favor different resolutions, the fitted component resolution and the resolution of an external label system should be reported separately.

Support selection uses the observed likelihood after target-preserving support-constrained refitting (Section 3.3.3). The corresponding BIC nominal dimension is a practical model-selection approximation, rather than an exact effective degrees of freedom result for a nonregular penalized mixture. Alternative degrees-of-freedom and EBIC rules are reported as sensitivity analyses in the Supplementary Material; no general selection-consistency result is claimed.

The fitting algorithm controls accepted generalized-M updates through smooth majorization and backtracking, with a separate step-halving safeguard during support-constrained refitting. Theorem 1 concerns the conditional convex subproblem, and Theorem 2 concerns an idealized sequence under explicit assumptions. The finite-tolerance fitted paths carry no claim of global optimality. Implementation and runtime diagnostics are provided in the Supplementary Material and are not statistical performance claims.

6.4. Limitations and future directions

E-CGL is intended for settings in which posterior-score contrast support is sparse or compressible. Weak separation, many weak active coordinates, or small samples can produce underselection, and mixture fitting remains sensitive to initialization and local optima. In the moderately dense $n = 300$ stress condition, the final single path selected 81.98 coordinates on average but attained only $F_{1,\eta} = 0.805$ and ARI 0.556. A selected support size close to the truth therefore did not eliminate the finite-sample difficulty created by diffuse signal. The main simulations fix the true K to isolate support recovery, while the separate K experiments provide diagnostics rather than an inferential guarantee. Real-data support cannot be evaluated by TPR_η or $F_{1,\eta}$ because its true coordinate support is unknown; the evidence instead rests on clustering performance, stability, and substantive interpretation. The Classic3 conclusions are also conditional on the fixed pretrained SPLADE revision and prespecified vocabulary filters. The selected support is coordinate-system dependent: token-level interpretation is meaningful because the representation is aligned with the vocabulary. It does not transfer unchanged to an arbitrarily rotated dense embedding space.

Further work is needed on effective degrees of freedom for centered penalized mixtures, joint uncertainty in K and support, and selection rules that remain stable under diffuse weak signals. The theory fixes K and d ; the high-dimensional and $d > n$ results are computational evidence rather than an asymptotic theorem. Adaptive weighting also remains sensitive to pilot contrasts and path coverage. Extensions that combine sparse posterior-score support with explicit dense-background components may also reduce the finite-sample cost observed in dense-support settings.

Centered natural-parameter group regularization defines an estimator of sparse posterior-score contrast support in vMF mixtures. Its target differs from a sparse component prototype: it retains coordinates that change posterior component comparisons while leaving common natural-parameter baselines unpenalized. The simulation and Classic3 analyses support this target-specific interpretation, and the dense-support stress experiments identify settings in which sparsification can be costly. E-CGL is therefore positioned as an estimator of posterior-score contrast support for sparse or compressible contrast structures, not as a uniformly superior replacement for dense or prototype-oriented mixture methods.

Appendix A. Proofs of theoretical results

The main text states the analytic bounds that determine each result; this appendix gives the complete arguments.

Proof of Lemma 1. The closed simplex and the product of the K closed Euclidean balls of radius $\kappa_{\max}$ are compact. Each vMF component density is positive and continuous on this set, so the observed log-likelihood is finite and continuous. The centered group penalty is also continuous. The criterion therefore attains its maximum by the extreme-value theorem. $\square$

Proof of Proposition 1. Taking the average of $v = \bar{v}\mathbf{1}_K + c$ and using $\mathbf{1}_K^\top c = 0$ determines $\bar{v}$ and hence c uniquely. The three zero-contrast statements are immediate. Finally, $\sum_{k < \ell} (v_k - v_\ell)^2 = K \sum_k (v_k - \bar{v})^2$, which gives the stated identity. $\square$

Proof of Proposition 2. Because $\kappa = \bar{\kappa}\mathbf{1}_K + h$ and $M = \mathbf{1}_K \bar{\mu}^\top + U$,

$$E = \text{diag}(\kappa)M = \bar{\kappa}\mathbf{1}_K \bar{\mu}^\top + \bar{\kappa}U + h\bar{\mu}^\top + \text{diag}(h)U.$$

Apply H_K , using $H_K\mathbf{1}_K = 0$, $H_KU = U$, and $H_Kh = h$. $\square$

Proof of Proposition 3. Because $P\mathbf{1}_K = \mathbf{1}_K$, P commutes with J_K and H_K . Permutation matrices are orthogonal and therefore preserve the Euclidean norm. Applying Proposition 4 gives equivariance of the proximal map. $\square$

Proof of Proposition 4. The spaces spanned by $\mathbf{1}_K$ and orthogonal to $\mathbf{1}_K$ are orthogonal. The penalty is zero on the first space and equals the Euclidean norm on the second. The minimization therefore preserves $J_K v$ and applies the standard group soft-thresholding map to $H_K v$. $\square$

Proof of Proposition 5. The vMF log-partition function has Hessian equal to the covariance matrix of a unit-norm random vector and is therefore positive semidefinite with operator norm at most one. The Hessian of the k th block of f_t consequently has operator norm at most N_k . The descent result follows by combining the smooth majorization inequality with optimality of the proximal update. $\square$

Proof of Theorem 1. For finite natural parameters, the vMF log-partition function is strictly convex. Its Hessian is the covariance matrix of a vMF random vector, and the vMF density is positive on the full sphere; hence

$\text{Var}(a^\top X) > 0$ for every nonzero $a \in \mathbb{R}^d$. Since $N_k > 0$, the sum of its K blocks is strictly convex. The standard large-argument expansion of the modified Bessel function gives

$$-\log C_d(\kappa) = \kappa - \frac{d-1}{2} \log \kappa + R_d(\kappa), \quad \kappa \rightarrow \infty.$$

Here $R_d(\kappa) = O(1)$ as $\kappa \rightarrow \infty$.

Consequently, as $\|\eta_k\|_2 \rightarrow \infty$,

$$\begin{aligned} N_k \{-\log C_d(\|\eta_k\|_2)\} - r_k^\top \eta_k &\geq (N_k - \|r_k\|_2) \|\eta_k\|_2 \\ &\quad - \frac{N_k(d-1)}{2} \log \|\eta_k\|_2 + N_k R_d(\|\eta_k\|_2). \end{aligned}$$

The positive linear coefficient implied by $\|r_k\|_2 < N_k$ dominates the logarithmic term. Thus f_t is coercive. Adding the finite convex group penalty preserves strict convexity and coercivity, so $F_t^{(a)}$ has a unique minimizer. If that minimizer is interior to the concentration bound, it is also the unique bounded-subproblem solution. The gradient of f_t is globally Lipschitz with constant no larger than $\max_k N_k$. Hence the initial trial step $1/\max_k N_k$ satisfies the exact majorization inequality and is bounded away from zero. Standard proximal-gradient theory yields convergence and the $O(u^{-1})$ objective rate (Beck and Teboulle, 2009). The strict inequality $\|r_k\|_2 < N_k$ excludes the degenerate case in which all observations receiving positive responsibility from a component have one common direction. The finite cap still gives existence in that case, but this theorem does not apply. $\square$

Proof of Proposition 6. The standard EM lower-bound identity expresses the observed log-likelihood difference as the auxiliary-function difference plus a nonnegative Kullback–Leibler divergence term. The penalty depends only on the parameters and can be subtracted from both sides, yielding the penalized result. $\square$

Proof of Theorem 2. Let $\Theta^{(t_j)} \rightarrow \bar{\Theta}$ be any convergent subsequence, which exists by compactness. Asymptotic regularity also gives $\Theta^{(t_j+1)} \rightarrow \bar{\Theta}$. After passage to a further subsequence if needed, boundedness of the step sizes gives $s_{t_j} \rightarrow \bar{s}$ for some $\bar{s} \in [\underline{s}, \bar{s}]$. Exact posterior responsibilities are continuous on the assumed compact interior set, so $\tau^{(t_j)} \rightarrow \tau(\bar{\Theta})$.

The gradient $\nabla f_\tau(E)$ is jointly continuous in (E, τ) on this set. For s bounded away from zero, the scaled proximal map is jointly continuous in its

input and in s . Moreover, $s_{t_j} \mathcal{G}_{s_{t_j}}(E^{(t_j+1)}; \tau^{(t_j)}) \rightarrow 0$. The vanishing natural-parameter residual therefore implies

$$\bar{E} = \text{prox}_{\bar{s}\lambda_\eta \mathcal{P}_a} \{\bar{E} - \bar{s} \nabla f_{\tau(\bar{\Theta})}(\bar{E})\}.$$

The proximal fixed-point characterization yields

$$0 \in \nabla f_{\tau(\bar{\Theta})}(\bar{E}) + \lambda_\eta \partial \mathcal{P}_a(\bar{E}).$$

At exact posterior responsibilities, Fisher’s identity for the observed-data score gives $\nabla f_{\tau(\bar{\Theta})}(\bar{E}) = -\nabla_E \ell(\bar{\Theta})$, proving the natural-parameter KKT inclusion. For the mixing block,

$$\pi^{(t_j+1)} - \frac{N(\tau^{(t_j)})}{n} \longrightarrow 0,$$

while $\pi^{(t_j+1)} \rightarrow \bar{\pi}$ and $N(\tau^{(t_j)}) \rightarrow N\{\tau(\bar{\Theta})\}$. Hence $\bar{\pi}_k = N_k\{\tau(\bar{\Theta})\}/n$. Because the limiting mixing proportions are strictly positive,

$$\frac{\partial \ell(\bar{\Theta})}{\partial \pi_k} = \frac{N_k\{\tau(\bar{\Theta})\}}{\bar{\pi}_k} = n$$

for every k , which is the interior simplex KKT condition with $\nu = n$. The objective, residuals, and KKT system are equivariant under any fixed component permutation. The argument does not apply to boundary or finite-tolerance cases excluded by the assumptions (Green, 1990; Wu, 1983; Tseng, 2004). $\square$

Data and code availability

The source corpora must be obtained from their original providers under the applicable terms. Derived document-level SPLADE matrices are not redistributed because redistribution permission for the source corpora has not been established. A redistribution-safe reproducibility archive containing the reviewed implementation, tests, preprocessing specifications, configuration files, random seeds and split records, exclusion records, vocabulary and payload checksums, compact numerical summaries, and result-generation scripts is available from the corresponding author on reasonable request. Large raw, path, cache, and private project files are not part of that archive.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex for English-language editing, translation, software-development assistance, and consistency checking. The authors reviewed and verified the resulting text, code, references, and numerical claims and take full responsibility for the content of the publication.

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